
Data-Driven Material Selection for Fire-Safe Spacecraft Design

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Abstract

Spacecraft fire safety requires material decisions under atmospheric, flow, geometric, and gravitational conditions for which direct testing is expensive and unevenly distributed. This report consolidates the project’s curated literature database and evaluates whether supervised machine learning can support two tasks: binary prediction of sustained ignition and regression of flame-spread rate (FSR). The evaluation deliberately separates interpolation among conditions represented within known papers from extrapolation to entire held-out papers. After deduplication, the ignition task contains 4,487 observations from 87 canonical papers; the FSR task contains 2,531 usable observations from 66 papers. Preprocessing, hyperparameter search, threshold selection, and augmentation are confined to training partitions.

For ignition, the selected row-level XGBoost model reaches a ROC–AUC of 0.9076 ± 0.0102 , whereas the selected unseen-paper XGBoost model reaches 0.7183 ± 0.0444 . For FSR, the selected random forest reaches $R^2 = 0.8790 \pm 0.0178$ and $\text{RMSE} = 0.0063 \pm 0.0005$ m/s for row-level interpolation; paper-grouped evaluation falls to $R^2 = 0.4576 \pm 0.2215$ and $\text{RMSE} = 0.0122 \pm 0.0041$ m/s. SHAP and permutation analyses identify oxygen, flow and transport quantities, thermal time scales, and fuel properties as important model inputs, but these associations are not interpreted causally. The central engineering result is therefore conditional: the database supports useful within-domain ranking and screening, but the generalization gap and source heterogeneity preclude using the models as stand-alone certification evidence. The models are positioned as traceable decision-support tools for prioritizing tests, identifying poorly covered regimes, and guiding database expansion.

Statement of scope and evidence

All numerical claims in this report are taken from version-controlled database validation files, frozen split metadata, model-selection records, and generated evaluation tables in the repository. The oral-defense deck is used to recover the intended narrative, not as the numerical source of record; it explicitly marks one metric page as not updated. Although the defense question mentioned extinction, the final repository contains completed pipelines only for ignition and FSR. Consequently, no extinction model or result is claimed here.

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Nomenclature

Symbol or acronym	Definition
c_p	Specific heat capacity, J/(kg K)
g	Acceleration level normalized by terrestrial gravity
k	Thermal conductivity, W/(m K)
α	Thermal diffusivity, m ² /s
ρ	Density, kg/m ³
FSR	Flame-spread rate, m/s
LOPO	Leave-one-paper-out
MAE	Mean absolute error
ML	Machine learning
MLP	Multilayer perceptron
OOD	Out of distribution
PR–AUC	Area under the precision–recall curve
RF	Random forest
RMSE	Root-mean-square error
ROC–AUC	Area under the receiver-operating-characteristic curve
SHAP	Shapley additive explanations
XGBoost	Extreme gradient-boosted decision trees

1 Introduction

1.1 Engineering context

Fire safety is a system-level requirement for a crewed spacecraft. A pressurized vehicle contains the three elements required for fire—oxidizer, combustible material, and potential ignition sources—within a confined volume from which evacuation, replacement of damaged equipment, and atmospheric clean-up may be difficult or impossible. Electrical systems are particularly important in this setting: they are indispensable to vehicle operation, yet faults, overheated components, and degraded insulation can supply ignition energy in close proximity to polymeric fuels. Spacecraft must also carry functional polymers, wiring, fabrics, packaging, personal items, and, for some mission concepts, polymer-rich radiation shielding. Material selection therefore cannot be reduced to eliminating every combustible item. It is an engineering compromise among fire performance, electrical and mechanical function, mass, manufacturability, human use, and mission architecture [1].

The historical record demonstrates why that compromise must be made explicitly. Guibaud et al. compile twelve acknowledged fire or smoke incidents from early crewed programmes through Mir. Their review includes the fatal Apollo 1 cabin fire, the Apollo 13 oxygen-tank event, smoke incidents aboard Salyut stations and Space Shuttles, and two fires aboard Mir. The incidents differ in initiating mechanism, atmosphere, detection, and consequence, but collectively expose recurring weaknesses in prevention, detection, crew response, suppression, and recovery. Three developed to a life-threatening scale, while several smaller events were controlled by removing electrical power before a larger fire formed. Importantly, dedicated detectors did not provide the initial warning in most of the minor events; crew members frequently noticed odour, smoke, or visible flame first [1]. These are findings of the cited historical review, not results generated by the present project.

Low-Earth-orbit operations have accumulated extensive experience, but a deep-space mission changes the consequences of a fire. The International Space Station can rely on established ground support, emergency return vehicles, and eventual resupply. A distant vehicle or surface habitat must remain safe when communication is delayed, rapid evacuation is unavailable, and consumed extinguishing agent or damaged hardware cannot promptly be replaced. Guibaud et al. further identify prospective low-pressure, oxygen-enriched exploration atmospheres, long exposure to radiation, intermediate lunar or Martian gravity, dust, waste accumulation, and increased operational autonomy as coupled fire-safety concerns [1]. The implication for materials engineering is direct: a material that is acceptable under a terrestrial screening configuration cannot be assumed to retain the same relative hazard under every spacecraft atmosphere, flow field, geometry, or gravity level.

1.2 Why spacecraft fire behaviour is a distinct prediction problem

Terrestrial intuition is incomplete in reduced gravity because buoyancy no longer organizes heat and mass transport in the familiar way. In normal gravity, density differences produced by heating generate a buoyant flow that affects oxygen delivery, flame shape, preheating of unburned fuel, smoke transport, and heat loss. In microgravity, forced ventilation and molecular diffusion become dominant transport mechanisms. Consequently, a weak imposed flow can have a large influence on whether a material ignites and on how rapidly a flame advances. Smoke can also migrate slowly or along ventilation paths rather than rising to a ceiling-mounted detector [1]. Reduced gravity does not simply multiply terrestrial burning behaviour by a constant correction factor; it changes the balance among transport processes.

The atmosphere introduces another set of coupled variables. Oxygen concentration affects the ability of a flame to be established and sustained, while total pressure changes gas density, transport, and cooling. Exploration atmospheres may depart from the terrestrial reference mixture of approximately 21% oxygen in nitrogen at 101.3 kPa. Guibaud et al. report that increasing oxygen content can make additional spacecraft materials flammable and increase subsequent fire growth, while low-pressure operation can reduce convective cooling of electrical equipment [1]. A safe material decision therefore depends on oxygen concentration and pressure together, not on a material name alone.

Geometry and material construction are equally important. The wire-flame literature synthesized by Rivera et al. shows that flame spread over insulated electrical wires depends on core and insulation materials, core diameter, insulation thickness, oxygen concentration, pressure, imposed flow, gravity, and spread direction. A conductive metal core can transfer heat ahead of a flame and preheat virgin insulation, but it can also remove heat from the burning region and act as a heat sink. Which effect dominates depends on the relative core and insulation dimensions, their thermophysical properties, and the surrounding conditions [4]. Melting and dripping introduce further regime changes. Thus “polyethylene-insulated wire”, for example, does not define a single fire response independently of its core, dimensions, orientation, and environment.

Two outputs are especially useful at the material-screening stage. The first is *ignition*, represented here as a

binary experimental outcome,

$$y_{\text{ign}} \in \{0, 1\}, \quad y_{\text{ign}} = \begin{cases} 1, & \text{ignition observed,} \\ 0, & \text{ignition not observed.} \end{cases} \quad (1.1)$$

The second is flame spread rate (FSR), the speed at which an established flame front advances over the fuel,

$$v_{\text{FSR}} = \frac{dx_f}{dt} \quad [\text{m s}^{-1}], \quad (1.2)$$

where x_f denotes a consistently defined flame-front position. Ignition addresses whether the imposed test and environmental conditions produce a flame; FSR quantifies one aspect of propagation after ignition. They are related but not interchangeable. A material-condition pair may be difficult to ignite yet support appreciable spread once ignited, or may ignite without yielding a steady, measurable spread rate. Treating the two as separate learning tasks preserves this distinction.

This report does *not* model flame extinction. Extinction is scientifically important and is discussed in the cited spacecraft-fire literature as part of the wider combustion problem [1], but it is outside the repository pipelines assessed here. The implemented scope is limited to classification of ignition and regression of FSR. Statements about extinction in the literature review are therefore contextual only and must not be read as an extinction prediction capability.

1.3 Motivation for a data-driven material-selection framework

The conventional spacecraft qualification process emphasizes controlled flammability tests and configuration-level hazard analysis. Guibaud et al. describe NASA’s material test philosophy, including a propagation test in which specimens are assessed in a prescribed worst-case atmosphere and are rejected if propagation exceeds the specified criterion or burning debris ignites an indicator material. They also show why a specimen-level pass/fail result cannot answer every system question: full-scale Apollo mock-up tests did not always reproduce the benign behaviour inferred from isolated-component tests, and gravity-dependent changes can alter both flammability and material ranking [1]. Physical testing remains indispensable, but a finite test matrix cannot densely cover all combinations of material properties, dimensions, oxygen content, pressure, flow, orientation, and gravity.

High-fidelity combustion simulations face a complementary difficulty. Flame spread couples solid heating and pyrolysis to gas-phase transport and reaction, often with melting, moving boundaries, radiation, and uncertain material properties. Rivera et al. note that analytical models necessarily simplify these processes and that computational models are difficult to validate over the full parameter space because reduced-gravity experiments are expensive and heterogeneous [4]. The objective of a data-driven model is not to remove the need for experiments or physics-based simulation. It is to learn cross-study relationships from measurements that already exist, quantify predictive performance under explicit validation rules, and support prioritization of the next material or condition to test.

The repository database makes such an analysis possible by placing literature observations into a common tabular representation. Each row associates a cited experiment with sample descriptors, fuel and core properties, atmospheric and flow variables, facility information, ignition method where available, and measured outputs. This structure naturally supports two supervised mappings. For a feature vector $\mathbf{x}$ describing the specimen and its environment,

$$\hat{p}_{\text{ign}} = f_{\text{cls}}(\mathbf{x}), \quad (1.3)$$

$$\hat{v}_{\text{FSR}} = f_{\text{reg}}(\mathbf{x}), \quad (1.4)$$

where f_{cls} estimates an ignition probability and f_{reg} estimates FSR in m s^{-1} . The predictions are conditional on the experimental domain represented in the database. They are not universal material constants.

Tabular scientific data have characteristics that matter for model choice: numerical and categorical predictors coexist, values may be missing, and the response may change across sparsely sampled regimes. In a broad comparison across eleven tabular datasets, Shwartz-Ziv and Armon found that XGBoost generally outperformed the deep tabular architectures examined and required less hyperparameter tuning; they also found that an ensemble combining XGBoost and deep models could outperform either family alone [5]. Their result is evidence for benchmarking strong tree ensembles, not a theorem that one algorithm must dominate every combustion dataset. Accordingly, this project treats model family as an empirical question and compares multiple candidates rather than assuming that architectural complexity guarantees better prediction.

Physics-informed machine learning provides a second motivation. Raissi, Perdikaris, and Karniadakis formulated physics-informed neural networks (PINNs) in which a neural approximation is trained against observed data while a differential-equation residual is penalized at collocation points [3]. Karniadakis et al. place such approaches within a broader framework in which observational, simulation, and physical information may be combined to address forward and inverse problems [2]. For a state u governed by

$$\frac{\partial u}{\partial t} + \mathcal{N}[u; \boldsymbol{\lambda}] = 0, \quad (1.5)$$

a canonical PINN loss has the form

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}}\mathcal{L}_{\text{residual}} + \lambda_{\text{bc}}\mathcal{L}_{\text{boundary}}, \quad (1.6)$$

where the additional terms constrain the learned field using governing equations and boundary information [3]. This idea is attractive for combustion data because physical knowledge could regularize learning where measurements are sparse. It also imposes a strict requirement: the chosen constraint must be valid for the target and variables being modeled. The present database consists principally of experiment-level tabular observations rather than resolved space–time fields with complete boundary conditions. Physics-informed ideas therefore motivate physically meaningful features and constraints, but they do not justify pretending that a conventional tabular predictor is a full PDE solver.

1.4 Research gap and question

The reviewed sources reveal a gap between spacecraft fire-safety needs and available predictive methods. The spacecraft literature documents the operational importance of ignition and spread under reduced buoyancy, as well as unresolved dependence on atmosphere, flow, geometry, and gravity [1]. The wire literature demonstrates that nonlinear machine-learning models can predict FSR from compiled experiments and that categorical regime information can materially improve random cross-validation performance [4]. General machine-learning literature supports tree ensembles as strong tabular baselines [5], while the PINN literature shows how physical equations can be embedded in learning when suitable field data and governing constraints are available [2, 3].

What remains insufficiently resolved is a unified, leakage-resistant evaluation of both fire initiation and propagation across heterogeneous literature sources, with validation that distinguishes interpolation among represented studies from transfer to held-out experimental sources or conditions. High random-split accuracy alone can reflect repeated or closely related configurations on both sides of a split. It does not establish that a model will generalize to a new paper, facility, material family, or environmental regime. Rivera et al. explicitly characterize their random ten-fold validation as robustness within their compiled wire dataset rather than strict transfer to a new source [4]. That limitation is central for engineering use, because material selection often asks precisely about a candidate outside the densest region of prior testing.

The research question guiding this report is therefore:

To what extent can supervised models trained on a curated, multi-source microgravity-combustion database predict (i) whether ignition occurs and (ii) the flame spread rate under spacecraft-relevant material, geometric, atmospheric, flow, and gravity conditions, while maintaining performance when evaluated beyond the experimental groups used for training?

The wording is deliberately conditional. “Predict” means estimate the two recorded outcomes within a stated feature domain and validation protocol. It does not mean certify flight hardware, recover an unobserved combustion mechanism, or model extinction.

1.5 Objectives, contributions, and scope

To answer the research question, the work is organized around four engineering objectives. First, it converts heterogeneous literature measurements into analysis-ready predictors with consistent SI units, explicit target definitions, and traceability to source publications. Second, it develops a dedicated ignition-classification workflow and a separate FSR-regression workflow. Third, it evaluates candidate models under data partitions designed to expose both within-domain interpolation and more demanding extrapolation or source-transfer behaviour. Fourth, it interprets the trained models only to the degree supported by the data, using feature-importance and error analyses as diagnostic tools rather than as proofs of causation.

The corresponding contribution is a reproducible decision-support framework linking a cited experimental database to two complementary predictions. The classifier provides a conditional ignition-risk estimate, and the regressor provides an estimated propagation speed when FSR is the relevant recorded output. Together they offer more information than a single undifferentiated “flammability” label. The framework also makes its limits visible: missing values, uneven source coverage, correlated measurements, and group-sensitive validation are treated as engineering constraints rather than hidden preprocessing details.

Table 1.1 states the boundary of the report. It prevents contextual discussion of the broader fire problem from being mistaken for implemented capability.

The intended use is screening and prioritization. A model may help identify combinations that warrant physical testing, compare candidate model classes, or reveal where the database is too sparse for confidence. Final material acceptance must still rest on applicable spacecraft standards, configuration analysis, and representative experiments. This distinction follows the lesson of the historical literature: safety depends not only on a specimen but on its atmosphere, installation, ignition sources, detection and mitigation provisions, and mission context [1]. Data-driven material selection is therefore best understood as one evidence layer in a larger fire-safety argument.

Table 1.1: Scope of the present data-driven study.

Included targets	Included conditioning information	Outside scope
Binary ignition outcome	Material and core properties; specimen geometry; oxygen concentration; pressure; gas and imposed-flow descriptors; gravity and facility descriptors where available	Flame extinction; smoke toxicity; heat-release rate prediction; suppression-system performance; full spacecraft fire dynamics
Flame spread rate in m s^{-1}	Literature-derived experimental conditions represented in the curated database	Flight qualification, causal certification, and unrestricted extrapolation to unrepresented materials or environments

2 Literature Review

2.1 Review strategy and source roles

This chapter synthesizes five repository-authoritative sources that address different layers of the research problem. Guibaud et al. provide the spacecraft fire-safety context, historical evidence, and deep-space challenges [1]. Rivera et al. review the physics of flame spread over electrical wires and present the closest prior machine-learning study to the present FSR task [4]. Schwartz-Ziv and Armon provide comparative evidence about model selection for tabular data [5]. Raissi, Perdikaris, and Karniadakis establish the PINN formulation [3], while Karniadakis et al. situate physics-informed learning within the wider scientific-machine-learning field [2]. The project’s oral-defense material informed the desired narrative order only; no numerical claim in this chapter is taken from that outline.

These source roles must remain distinct. Historical incident counts, spacecraft atmosphere descriptions, and deep-space concerns are attributed to Guibaud et al. Physical and machine-learning statements specific to wire flame spread are attributed to Rivera et al. Claims about tabular algorithm comparisons are limited to the datasets and protocol of Schwartz-Ziv and Armon. PINN equations describe methods reported in the PINN literature; they do not imply that this project’s tabular pipelines solved a combustion PDE. Finally, none of the performance values reported for earlier studies is presented as a result of this project.

Table 2.1: Authoritative sources and their use in this report.

Source	Principal evidence used	Limitation for the present question
Guibaud et al. [1]	Spacecraft incidents, fire-safety strategy, reduced-buoyancy implications, and deep-space challenges	Review and risk context; not a trained ignition or FSR predictor
Rivera et al. [4]	Wire-flame physics, heterogeneous experimental predictors, categorization, augmentation, and ML FSR benchmarks	Focuses on wire FSR; random cross-validation is not strict held-source-out transfer
Shwartz-Ziv and Armon [5]	Controlled comparison of XGBoost with deep models on eleven tabular datasets	General tabular evidence, not combustion-specific evidence
Raissi et al. [3]	Differential-equation residuals, collocation points, and forward/inverse PINN formulation	Assumes suitable governing equations and coordinate-resolved information
Karniadakis et al. [2]	Broader synthesis of physics-informed machine learning and data-physics integration	Research framework rather than a spacecraft-material qualification method

2.2 Spacecraft fire safety as a material-selection problem

2.2.1 Lessons from incidents and evolving practice

The history reviewed by Guibaud et al. shows that spacecraft fire safety evolved through incident investigation, atmospheric redesign, material screening, detection hardware, suppression equipment, and crew procedures. Their chronology identifies twelve acknowledged fire or smoke events. Apollo 1 remains the clearest demonstration of coupled design hazards: an oxygen-rich cabin, combustible interior materials, possible electrical ignition, rapid pressure rise, and an inward-opening hatch combined in a fatal accident during a ground test. The precise ignition source was not established, but subsequent actions included atmospheric changes, removal or protection of combustibles, and extensive component and mock-up flammability testing [1].

Apollo 1 also demonstrated that material behaviour cannot always be inferred from an isolated specimen. Guibaud et al. report that components showing acceptable individual behaviour did not necessarily prevent excessive propagation after integration into a full spacecraft mock-up. The configuration changed the available fuel paths, heat feedback, and interactions among adjacent items. This observation remains relevant to data-driven screening: a row-level material prediction can inform a decision, but cannot by itself represent every installation detail in a complete vehicle [1].

Later incidents shifted attention toward electrical equipment and operational response. The Salyut and Shuttle records include failed fans, short circuits, overheated resistors, fused wires, and failed electronic components. In the five Shuttle events summarized by Guibaud et al., the crew interrupted power before a full-scale fire developed, while onboard detection systems did not provide the decisive first warning. The 1997 Mir fire showed a different failure mode: a lithium-perchlorate oxygen canister produced an energetic flame and dense smoke, obstructed access

to one escape vehicle, and could not be directly suppressed by the available water-based foam. Crew members had to improvise while contending with visibility, equipment access, and respirator problems [1]. The literature’s point is not that every spacecraft fire resembles a wire fire, but that ignition source, material, atmosphere, access, and mitigation interact at system scale.

Present practice uses multiple defence layers. Guibaud et al. describe fire-resistant material selection, configuration analysis, redundant or distributed smoke detection, electrical and ventilation shutdown, portable extinguishers, atmospheric monitoring, crew masks, and rehearsed procedures on the International Space Station. This layered architecture has benefited from earlier programmes, yet it still relies on knowledge of material response and combustion signatures in low gravity. The review argues that uncertainties remain in translating terrestrial test performance to reduced gravity and in demonstrating suppression performance for every relevant scenario [1].

2.2.2 Qualification tests and their limits

Spacecraft material control is intentionally conservative. The test history summarized by Guibaud et al. includes NASA’s vertical flame-propagation screening, in which multiple specimens are exposed to an ignition source under a specified atmosphere. The cited criterion considers both propagation distance and ignition by burning debris. At the vehicle level, a material that does not pass an isolated test may still require configuration analysis when mission-critical function prevents simple removal. Separation, containment, barriers, and elimination of propagation paths become part of the safety argument [1].

This approach is essential but does not produce a continuous map over all operating conditions. A pass/fail test answers a deliberately bounded qualification question. It does not directly estimate a probability of ignition for every pressure and oxygen concentration, nor does it yield FSR over arbitrary flow and gravity. Furthermore, Guibaud et al. cite reduced-gravity observations in which limiting behaviour differs from normal-gravity expectations and describe partial-gravity studies where relative material ranking can change [1]. A data-driven surrogate is useful precisely in this space between discrete experiments: it can organize existing observations, interpolate under controlled assumptions, and expose regions where direct testing remains necessary. It cannot supersede the qualification test from which the safety requirement is defined.

2.3 Ignition and flame spread under reduced buoyancy

2.3.1 Coupled solid- and gas-phase processes

Ignition is a transition rather than an intrinsic label attached permanently to a material. Energy supplied by an igniter or fault heats the solid; heat is stored and conducted through the specimen and, if the material decomposes, volatile fuel is generated. Gas-phase transport then mixes fuel and oxidizer, while losses to the core, surroundings, and unheated material compete with heat feedback. The literature reviewed by Rivera et al. emphasizes that even reduced analytical descriptions require material properties, dimensions, gas properties, and heat-flux estimates that are not uniformly reported across experiments [4]. Consequently, an observed yes/no ignition outcome is conditional on the entire experimental configuration.

A useful scale for solid heating is the thermal diffusion time

$$t_{\text{diff}} \sim \frac{\ell_c^2}{\alpha_s}, \quad \alpha_s = \frac{k_s}{\rho_s c_{p,s}}, \quad (2.1)$$

where ℓ_c is a characteristic solid length, k_s is thermal conductivity in $\text{W m}^{-1} \text{K}^{-1}$, ρ_s is density in kg m^{-3} , $c_{p,s}$ is specific heat in $\text{J kg}^{-1} \text{K}^{-1}$, and α_s is thermal diffusivity in $\text{m}^2 \text{s}^{-1}$. This scaling does not determine ignition by itself. It shows why thickness and thermophysical properties jointly affect the time over which a specimen responds to heating, and why separately recorded database columns can have interacting rather than additive effects.

After ignition, flame spread requires successive heating and pyrolysis of virgin material. Rivera et al. reproduce an analytical opposed-flow flame-spread relation in which spread speed depends on heated length, solid thermal inertia, pyrolysis and ambient temperatures, several flame and external heat-flux terms, radiative loss, and a kinetic contribution [4]. At the level needed here, that balance can be written schematically as

$$v_{\text{FSR}} \sim \frac{\dot{q}_{\text{net}}'' \ell_h}{\rho_s c_{p,s} \delta (T_p - T_0)}, \quad (2.2)$$

where $\dot{q}_{\text{net}}''$ denotes net heating of the unburned solid, ℓ_h is a heated length, δ is a solid thickness scale, T_p is a pyrolysis-temperature scale, and T_0 is the initial temperature. Equation (2.2) is a qualitative energy-balance representation, not the regression equation used by this project. It makes the key coupling explicit: any variable that changes heat feedback, heat storage, or heat loss can alter spread speed.

2.3.2 Microgravity transport

On Earth, gravity-induced buoyancy contributes a characteristic velocity and organizes the flame and smoke plume. In microgravity that contribution is suppressed, so imposed ventilation and diffusion control transport to a much

greater degree. Guibaud et al. connect this change to flammability, spread, smoke movement, detector response, and suppression uncertainty. A fire may produce combustion products that take longer to reach a sensor, while a relatively small ventilation flow can reshape a non-buoyant flame [1]. This is why “zero gravity” cannot be represented only as one more categorical label while all other transport variables are ignored.

Dimensionless groups help organize the competing mechanisms. For characteristic velocity U , length L , gas kinematic viscosity ν , and gas thermal diffusivity α_g ,

$$\text{Re} = \frac{UL}{\nu}, \quad \text{Pe} = \frac{UL}{\alpha_g}, \quad \text{Pr} = \frac{\nu}{\alpha_g}. \quad (2.3)$$

These groups compare inertial to viscous transport, advection to thermal diffusion, and momentum to thermal diffusivity, respectively. Their value in a heterogeneous database is representational: experiments with different dimensions and velocities may occupy related transport regimes. They do not remove the need to preserve dimensional predictors or account for geometry.

Opposed and concurrent flow are physically different configurations. In opposed spread, gas flow is directed toward the advancing flame front; in concurrent spread, flow and flame advance share a direction. Rivera et al. identify spread direction as an important categorical descriptor and report that adding such regime information substantially improved predictions in their compiled wire dataset [4]. The result supports explicit representation of configuration, but it must be interpreted within that study’s cross-validation protocol. It does not prove that a direction label captures all flow physics.

Partial gravity adds further complexity. Guibaud et al. review experiments and modelling that indicate non-monotonic responses for some fire quantities between microgravity and terrestrial gravity. They report different trends for upward and downward propagation and note that material ranking may change between normal, lunar, and Martian gravity [1]. The engineering consequence is that a predictor should not silently infer partial-gravity behaviour from an arbitrary linear interpolation between microgravity and terrestrial gravity. Sparse coverage must instead be treated as an extrapolation problem and reported as such.

2.3.3 Atmosphere, pressure, and oxygen

Oxygen concentration and pressure appear repeatedly in the spacecraft literature because mission atmosphere is both a life-support and structural design choice. Guibaud et al. describe the pure-oxygen, low-pressure atmosphere used in early American flight and the much higher pressure pure-oxygen condition present during the Apollo 1 ground test. Later spacecraft generally adopted nitrogen-diluted atmospheres, while proposed exploration vehicles may again use reduced total pressure and elevated oxygen fraction to address structural mass and extravehicular-activity considerations [1]. Fire response must therefore be considered over a domain, not only at terrestrial sea-level conditions.

Oxygen fraction is not equivalent to oxygen partial pressure, and neither alone captures all pressure effects. For an ideal mixture the oxygen partial pressure is

$$p_{\text{O}_2} = X_{\text{O}_2} P, \quad (2.4)$$

with mole fraction X_{O_2} and total pressure P in pascals. Equal values of p_{O_2} can still coexist with different total gas density and transport properties when diluent and total pressure differ. Rivera et al. consequently include oxygen concentration and pressure as separate predictors in the wire FSR problem [4]. This supports a multivariate representation rather than collapse to a single atmosphere variable.

2.4 Electrical-wire flame spread

Electrical wires are a particularly relevant material system because conductor and insulation form a coupled thermal geometry. Rivera et al. synthesize experiments using laboratory wires with metal cores and polymer insulation. Their review identifies core material and diameter, insulation material and thickness, oxygen concentration, pressure, forced flow, gravity, and orientation as influential quantities [4]. These variables also illustrate why the broader database is heterogeneous: a nominal material may appear in different specimen forms, and the same environment may produce different outcomes for different length scales.

The central conductor has a dual role. A high-conductivity core can move heat ahead of the flame and assist preheating. The same core can drain energy away from the reaction and pyrolysis region, especially when its cross-section is large relative to the insulation. Rivera et al. attribute this balance to core and insulation dimensions and properties, environmental conditions, and gravity [4]. A monotonic rule such as “higher core conductivity always increases FSR” would therefore contradict the reviewed physics. Flexible nonlinear models are attractive because they can represent conditional interactions, but flexibility also increases the need for careful validation.

Insulation behaviour introduces discrete regimes. Polymer type affects thermophysical response, while melting and dripping can change fuel geometry and energy transport. Orientation determines whether gravity assists movement of molten insulation in normal or partial gravity. Rivera et al. therefore encode core material, insulation material, spread direction, and configuration as categorical predictors in addition to numerical properties [4]. This

is an important methodological lesson: a table containing only bulk conductivity, density, and heat capacity may omit morphological or chemical distinctions that those properties do not fully identify.

The wire literature also exposes a transferability issue. Studies may concentrate on one polymer, core, diameter range, apparatus, pressure regime, or flow direction. When observations from the same study are divided randomly among training and test folds, the test set can remain close to the training distribution. Rivera et al. acknowledge that their random ten-fold cross-validation measures robustness within the compiled literature dataset rather than strict transfer to an entirely new source [4]. For material selection, this distinction is critical because a new candidate is often associated with a new material family or experimental campaign.

2.5 Machine learning for flame-spread prediction

2.5.1 Closest prior work

Rivera et al. provide the closest direct precedent for data-driven FSR prediction. They compile approximately 1200 wire-flame observations and describe numerical features for core, insulation, geometry, atmosphere, flow, and gravity, together with categorical indicators for material and spread configuration. They compare linear regression, decision tree, k -nearest neighbours, elastic net, stochastic gradient descent, Bayesian ridge, gradient boosting, support-vector regression, kernel ridge, and a multilayer perceptron [4]. The breadth of this benchmark is useful because it avoids treating one model family as the default.

Their baseline using numerical predictors alone produced weaker performance than versions that included one-hot encoded categorical variables. Under categorized random ten-fold cross-validation, several nonlinear models attained mean coefficients of determination around or above 0.90; the exact values vary by model and augmentation strategy [4]. These are literature results. They cannot be transferred directly to this project’s database, targets, partitions, or code, and they do not establish held-paper generalization.

Rivera et al. also compare bootstrap resampling, small numerical perturbations, and a generative adversarial network (GAN). Bootstrap augmentation was the most consistently beneficial for tree- and neighbourhood-based models in their reported experiments. The authors stress that resampling preserves observed combinations, whereas noise and GAN generation can produce statistically plausible combinations without guaranteeing geometric consistency, conservation, or other physical constraints [4]. This qualification matters: synthetic tabular rows are not new experiments. If material density, conductivity, dimensions, and categorical identity are perturbed independently, the resulting row may not correspond to a realizable specimen.

The prior study reports a small number of negative FSR predictions from unconstrained regressors and correctly identifies them as physically inadmissible [4]. This motivates non-negative output handling, target transformation, or explicit post-fit diagnostics. More generally, machine-learning performance must be judged against physical admissibility as well as aggregate error. A model can obtain a favourable mean score while failing in the low-FSR region that is most relevant to a safety boundary.

2.5.2 Model choice for tabular data

The structure of the combustion database resembles the broad statistical category studied by Shwartz-Ziv and Armon: fixed columns, heterogeneous units, mixed numerical and categorical features, missingness, and no spatial or sequential locality comparable to pixels or words. Their controlled study evaluates four deep tabular architectures and XGBoost across eleven classification and regression datasets, with a common hyperparameter-optimization protocol. XGBoost outperformed the deep models on most of the considered datasets, particularly those not used in the original deep-model papers, and its tuning converged more readily in their experiments [5].

The conclusion is narrower than “deep learning never works for tables.” Shwartz-Ziv and Armon explicitly invoke the no-free-lunch principle and report that an ensemble containing both deep models and XGBoost achieved the best average relative performance in their comparison [5]. For this project, the defensible inference is to include gradient-boosted trees as a strong baseline and compare them fairly with other candidates. Model selection must follow held-out evidence under the relevant engineering split, not reputation or architectural novelty.

Tree ensembles are plausible for this task because thresholded partitions can capture nonlinear interactions and accommodate regime changes, while neural and kernel models offer different smoothness and representation biases. No family removes the need for leakage control. Imputation, scaling, categorical encoding, feature selection, and hyperparameter optimization must be fitted using training data only. Repeated measurements and paper-specific clusters should remain grouped where the evaluation question concerns transfer to a new source. These are methodological requirements of the present project, not performance claims borrowed from the cited tabular benchmark.

2.6 Physics-informed machine learning

2.6.1 PINN formulation

Raissi et al. define PINNs for systems represented by nonlinear differential equations. For a latent field $u(t, \mathbf{x})$,

$$\frac{\partial u}{\partial t} + \mathcal{N}[u; \boldsymbol{\lambda}] = 0, \quad \mathbf{x} \in \Omega, \quad t \in [0, T], \quad (2.5)$$

the neural network approximates u , and automatic differentiation evaluates the residual

$$r_\theta(t, \mathbf{x}) = \frac{\partial u_\theta}{\partial t} + \mathcal{N}[u_\theta; \boldsymbol{\lambda}]. \quad (2.6)$$

Training combines mismatch to available observations and initial or boundary data with residual error at collocation points,

$$\mathcal{L}(\theta) = \frac{1}{N_u} \sum_{i=1}^{N_u} |u_\theta(t_i, \mathbf{x}_i) - u_i|^2 + \frac{1}{N_r} \sum_{j=1}^{N_r} |r_\theta(t_j, \mathbf{x}_j)|^2. \quad (2.7)$$

The framework supports forward inference with known physical parameters and inverse identification in which parameters in $\mathcal{N}$ are learned from observations [3].

Raissi et al. demonstrate the formulation on nonlinear Schrödinger, Allen–Cahn, Navier–Stokes, Korteweg–de Vries, and Burgers equations, using continuous- and discrete-time variants. They emphasize that PINNs complement rather than replace mature numerical methods and identify unresolved issues involving architecture, optimization, convergence, noise, and uncertainty [3]. Karniadakis et al. broaden the discussion to scientific settings where observational data may be scarce, noisy, or incomplete and prior physical knowledge can constrain an otherwise large hypothesis space [2].

2.6.2 Relevance and limitation for combustion tables

The appeal to spacecraft combustion is evident. Reduced-gravity experiments are costly, and conservation or transport relations contain information not present in a small set of labels. A valid physics-informed constraint could discourage impossible predictions and improve data efficiency. Dimensionless groups, positivity, energy-balance trends, and known symmetries are possible levels of physical structure, ranging from feature engineering to hard or soft constraints. The PINN literature provides a principled reason to investigate such structure [2, 3].

However, the standard PINN formulation requires more than a list of experiment-level rows. It needs a specified state field, spatial and temporal coordinates, a governing differential operator, and sufficient initial or boundary information for the target problem. The curated database records outcomes such as ignition and FSR under varied apparatuses; it is not a common mesh of temperature, species concentration, velocity, and reaction-rate fields. Applying Equation (2.7) without a defensible operator and boundary conditions would create the appearance of physics without its mathematical content.

This limitation is reinforced by Rivera et al., who conclude that the theoretical framework for their heterogeneous wire data remains incomplete for direct use of fully physics-informed models [4]. The scientifically appropriate hierarchy is therefore: preserve SI units and physically meaningful variables; encode known regimes; enforce indisputable output constraints such as non-negative FSR; evaluate transfer across sources; and reserve PDE-residual PINNs for datasets and subproblems where the governing fields and boundaries are actually available. This project can be physics-aware without claiming a PINN where none was implemented.

2.7 Synthesis and explicit research gap

The literature converges on four findings. First, spacecraft fire is a credible system hazard whose consequence increases when evacuation, resupply, and recovery are restricted [1]. Second, ignition and spread depend jointly on material, geometry, atmosphere, flow, and gravity; reduced buoyancy changes transport rather than merely slowing or accelerating a terrestrial flame by a fixed factor [1, 4]. Third, nonlinear supervised models can recover useful relationships from heterogeneous wire-flame tables, especially when physically meaningful categories are retained [4]. Fourth, strong tabular baselines and physical constraints should be evaluated empirically and applied only within the assumptions their source literature supports [2, 3, 5].

Three gaps remain. The *target gap* is that the closest ML study concentrates on FSR, whereas an engineering screening workflow also needs an explicit ignition model. The *validation gap* is that random cross-validation within a compiled literature table does not measure transfer to a held-out paper, apparatus, or condition regime. The *integration gap* is that broad spacecraft studies establish operational relevance while wire and ML studies establish local predictive feasibility, but the evidence has not yet been organized into one traceable framework that predicts both ignition and FSR under spacecraft-relevant conditions without overstating physical completeness.

The present work addresses these gaps by constructing separate classification and regression tasks from the repository database, comparing model families appropriate to mixed tabular data, and distinguishing interpolation from more demanding grouped or extrapolative evaluation. Its claims are intentionally bounded. The project predicts recorded ignition and FSR outcomes; it does not model extinction, certify flight materials, or solve the full reacting-flow problem. That boundary is not a weakness in presentation but a consequence of the available evidence. A reliable material-selection tool must make clear both what has been learned from existing tests and where a new experiment remains the only defensible next step.

3 Methodology

3.1 Study design and prediction tasks

This work treats the curated microgravity-combustion database as an observational, multi-study data set rather than as a collection of independent and identically distributed measurements. An experimental paper commonly contributes a series of closely related rows obtained with the same apparatus, material preparation, diagnostic practice, and range of operating conditions. Consequently, two rows from one paper can be substantially more similar than two rows drawn from different papers even when their tabulated physical variables overlap. The methodology was therefore organized around two distinct prediction tasks. The first was binary classification of whether ignition occurred. The second was regression of flame spread rate (FSR), expressed in m s^{-1} . FSR was modeled only where a numeric FSR observation was present. No extinction target or extinction model was defined.

For each task, interpolation and extrapolation were evaluated separately. *Row interpolation* asks how accurately a model predicts a held-out row when other rows from the same paper may remain in training. It is useful for estimating performance within the support of campaigns already represented in the database, but it does not test transfer to a new publication. *Paper-grouped extrapolation* instead holds out complete canonical papers, so that no paper identity occurs in both training and test partitions. It asks whether relationships learned from the available literature transfer to a previously unseen paper or campaign. The grouped estimate is the primary evidence for cross-paper generalization. The deliberate use of both protocols makes the otherwise hidden effect of campaign dependence measurable.

The two pipelines followed the same high-level sequence: load and standardize the data; freeze all outer partitions; perform hyperparameter search within each outer training partition; fit the selected configuration to that partition; evaluate it once on the untouched outer test partition; aggregate persisted predictions and metrics; choose separate interpolation and extrapolation champions by a declared policy; and only then refit deployable models on all labeled rows. Evaluation models and full-data refits have different inferential roles. The former provide out-of-sample comparison evidence, while the latter are operational artifacts and do not supply unbiased performance estimates.

3.2 Database ingestion and canonicalization

3.2.1 Source structure and evidenced standardization

Both analyses used the task-local `Microgravity_Database.csv`. The CSV contains a category banner in its first row, a field-name header in its second row, and observations thereafter. The loaders therefore read the CSV after skipping the category row and attempted UTF-8, UTF-8 with byte-order mark, CP1252, and Latin-1 encodings. The persisted validation reports identify CP1252 as the successful encoding for both analyzed files. The FSR loader additionally accepts Excel workbooks with one- or two-level headers, but the persisted analysis reported the CSV parse `skiprows=1; encoding=cp1252`. These operations describe schema ingestion and computational standardization; the repository does not document enough of the upstream literature-search or manual transcription process to justify broader claims about database construction.

Whitespace and placeholder tokens were normalized, duplicate headers and all-empty columns were removed where applicable, and numeric values were parsed from unit-bearing cells. The ignition pipeline implemented explicit conversion for sample dimensions to metres, pressure to pascals, and flow velocity to m s^{-1} , using either a recognized cell suffix or the unit in the column header. Gravity was represented relative to terrestrial gravity, g/g_{Earth} . A microgravity text token without a numeric value was mapped to 10^{-6} , while acceleration values in m s^{-2} , cm s^{-2} , or mm s^{-2} were converted to this relative scale. Ignition-power entries describing heat flux, current, or other quantities that were not watts were not coerced into power values. In the FSR pipeline, columns were classified as numeric when at least 60% of their available cells could be parsed from a leading number. Restricting parsing to a leading numeric token prevents chemical labels such as N_2 or CO_2 from being mistaken for measurements. Thus, for example, “101.3 kPa” is numeric while “N2” remains categorical.

Six fields were dropped before modeling in both tasks: material of sample, rig name, internal geometry, internal dimensions, facility, and free-form information. Paper identifiers and bibliographic fingerprints were never supplied as predictors. For ignition, the post-outcome columns flame length, FSR, heat-release rate (HRR), and smoke/aerosols were explicitly excluded. For FSR, the selected FSR target, duplicate FSR-like fields, citation and author identifiers, notes, and detected post-outcome fields such as flame length, HRR, smoke/aerosols, and extinction-like fields were excluded. This exclusion is a leakage control, not an extinction analysis.

3.2.2 Canonical paper identity

The grouping unit was a canonical paper identifier. A valid DOI was normalized by removing URL or `doi:` prefixes, lowercasing, and retaining the canonical DOI expression. The resulting identifier had the form

doi::<normalized-doi>. If no valid DOI was available, the citation was lowercased, punctuation was removed, whitespace was collapsed, and the identifier `citation::<normalized-citation>` was used. Raw DOI strings were therefore not used directly as groups. This canonicalization is essential: cosmetic differences in citation formatting must not permit one publication to appear under multiple group labels and thereby cross a grouped train-test boundary.

3.2.3 Task-specific cohorts

The ignition target was the database field `Ignition`, mapped to $y_i \in \{0, 1\}$ for No and Yes. Training required every target to be recognized. Rows were deduplicated on the complete engineered feature vector, target, and canonical paper identifier, retaining the first occurrence. The persisted `fable-data-v5` cohort contains 4,487 observations from 87 papers after 570 duplicates were removed. There were 3,383 ignition-positive and 1,104 ignition-negative rows, corresponding to a positive prevalence of 0.75396. Twenty-nine blank source rows were also identified in the validation report.

For regression, the loader selected the FSR-like column with the greatest number of numeric values; the persisted target was `FSR (m/s)`. Rows lacking numeric FSR were removed, and duplicates were counted after target filtering. The `fsr-data-v2` cohort contains 2,531 usable rows from 66 papers, derived from 5,057 raw rows: 2,467 rows lacked FSR and 59 duplicates were removed. Papers contributed between 1 and 193 usable rows (median 22.5, mean 38.35), demonstrating substantial imbalance in campaign size. The observed FSR ranged from 0 to 0.211 m s^{-1} , with mean 0.008774 m s^{-1} and standard deviation 0.017252 m s^{-1} .

Table 3.1: Persisted analysis cohorts. Counts are those recorded by the task-specific validation reports after target filtering and deduplication.

Task	Raw rows	Analyzed rows	Papers	Duplicates removed
Ignition classification	5,086	4,487	87	570
FSR regression	5,057	2,531	66	59

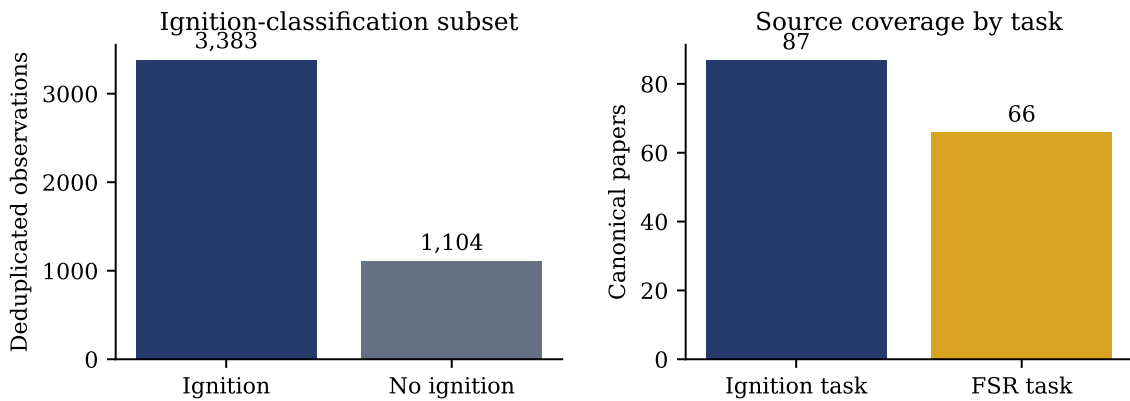


Figure 3.1: Task-specific database composition after the preprocessing described in this chapter. Counts are generated directly from the persisted validation summaries; the two tasks use different target-availability rules and therefore different paper cohorts.

3.3 Predictor representation and preprocessing

3.3.1 Ignition feature engineering

Ignition classification used a fixed manifest of 32 numeric and five categorical features. This manifest was invariant across all candidates. Fuel properties comprised density (kg m^{-3}), thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$), specific heat ($\text{J kg}^{-1} \text{K}^{-1}$), pyrolysis temperature (K), thermal diffusivity ($\text{m}^2 \text{s}^{-1}$), and volumetric heat capacity ($\text{J m}^{-3} \text{K}^{-1}$). Corresponding core properties included density, thermal conductivity, specific heat, and volumetric heat capacity.

Gas and transport variables were oxygen concentration, gas molar mass, gas specific heat, gas thermal conductivity, gas density, gas thermal diffusivity, kinematic viscosity, Reynolds number, Péclet number, Prandtl number, and thermal diffusion time in seconds. Operating conditions comprised pressure in pascals, signed flow velocity and absolute flow speed in m s^{-1} , relative gravity, ignition power in watts, and ignition duration in seconds. Geometry was represented by half-thickness, a characteristic length, and as many as three parsed sample dimensions, all in metres.

The five categorical variables were engineered from the schema: sample geometry (wire, flat, cylindrical, spherical, or residual category), diluent, signed-flow direction (coflow, counterflow, or quiescent), gravity regime (microgravity, partial gravity, Earth gravity, or hypergravity/unknown), and ignition method (open flame, radiative heater, discharge, wire/coil, or unknown). Signed flow and absolute flow were intentionally both retained because direction and magnitude encode different information. Derived quantities already present in the database, including Reynolds, Péclet, and Prandtl numbers and thermal diffusion time, were treated as predictors; their inclusion creates correlated representations and must be considered when interpreting importance.

Missingness was heterogeneous. Flow variables were missing in 897 ignition rows, gravity in 54, and gas-derived transport variables in 48. Ignition power was missing in 3,606 rows and ignition time in 2,400; the third sample dimension was missing in 2,049. By contrast, many tabulated material properties had no recorded missing values. These patterns mean that apparatus-related variables can partly encode which campaigns measured or reported them, even though explicit paper identifiers were excluded.

3.3.2 FSR feature manifest

FSR regression used a detected manifest rather than the ignition pipeline’s renamed fixed manifest. The persisted manifest contains 29 numeric and four categorical fields. Numeric fields cover the sample-dimension string parsed numerically, half-thickness and characteristic length; fuel and core thermophysical properties; oxygen concentration; gas molar mass, heat capacity, conductivity, density, thermal diffusivity, and kinematic viscosity; Reynolds, Péclet, and Prandtl numbers; thermal diffusion time; pressure in pascals; flow velocity in ms^{-1} ; relative gravity; and ignition power and time. Geometry, diluent, ignition method, and ignition outcome are categorical. The manifest tags ignition power, ignition time, ignition method, and ignition outcome as apparatus features and the remainder as physics features. However, role tags were retained for reporting only: all candidates used the same full feature list, so the `feature_set=all` designation is substantive and no physics-only feature ablation was performed.

The FSR target is in ms^{-1} , and names carrying SI units were interpreted accordingly. The generic FSR parser extracts leading numbers but does not establish additional unit conversions beyond the recorded schema; no stronger standardization claim is made. Missingness again concentrated in apparatus and flow fields: among analyzed rows, flow was missing in 709, ignition power in 2,198, ignition time in 1,725, gravity in 30, and several gas/transport variables in 33.

3.3.3 Fold-local transformations

All learned preprocessing was fitted on training data only. For scikit-learn-style models, numeric missing values were replaced by the training-fold median. XGBoost retained native missing values. Numeric scaling by a training-fold `StandardScaler` was applied for distance-, margin-, and neural-network families, but not for tree models. Categorical missingness was replaced by the constant “Unknown” and categories were one-hot encoded with unknown-category-safe handling. This arrangement prevents test-set medians or category vocabularies from entering model fitting.

3.4 Candidate models, weighting, and augmentation

The ignition configuration declared 26 candidates spanning XGBoost, decision trees, k -nearest neighbors (KNN), multilayer perceptrons (MLP), and support-vector machines (SVM). Candidate recipes varied paper weighting, balanced class weighting, focal versus logistic XGBoost objectives, distance-weighted KNN, and linear versus radial-basis SVM. The FSR configuration declared 34 candidates: 10 XGBoost, seven random-forest, six KNN, five MLP, and six support-vector regression (SVR) recipes. Regression candidates varied paper weighting, tree loss, signed-log($1 + y$) target transformation, KNN distance weighting, SVR kernel, and XGBoost paper bagging. Predictions from a transformed-target candidate were inverse-transformed to original FSR units before evaluation.

The persisted comparison pools were smaller than these configured portfolios. The merged FSR manifest contains 25 candidates (50 candidate–augmentation conditions), and the formal ignition selection record omits MLP candidates even though MLP shard metrics are present. Results and champion claims in this report therefore refer to the candidates actually represented in the persisted leaderboards and selection records, not to an assumed complete execution of every configured recipe.

Campaign-size weighting was defined from the number n_p of training rows in paper p . The principal choices were

$$w_{ip}^{(\text{none})} = 1, \quad w_{ip}^{(\text{sqr})} = n_p^{-1/2}, \quad w_{ip}^{(\text{inverse})} = n_p^{-1}. \quad (3.1)$$

For ignition, optional balanced class weights were multiplied by the paper weights and the result renormalized. Families with native sample-weight support received these values during fitting. Weight-blind families used deterministic probability-proportional resampling so that a weighting option was not silently ignored. The FSR MLP minimized weighted mean squared error; FSR KNN and SVR used the analogous resampling path.

Every FSR candidate was evaluated under two training-data conditions. `rd` used the real outer-training rows. `rd_bt` appended 1,000 rows sampled with replacement from those training rows; each copy retained its source paper

identifier. This bootstrap operation changes the empirical frequency of observed conditions. It neither generates independent experiments nor expands physical support, and the test set was never augmented. Candidate and augmentation together formed the regression selection unit.

3.5 Frozen outer evaluation protocols

The ignition split file used base seed 42 and three consecutive seeds. For each seed, it stored a stratified 80/20 row holdout, a five-fold stratified row cross-validation, and a five-fold stratified grouped cross-validation. It also stored 87 leave-one-paper-out (LOPO) folds, one for each canonical paper. Thus, the primary interpolation and grouped protocols each comprised 15 outer folds, the supplementary holdout comprised three, and LOPO supplied a paper-level robustness analysis. Split integrity was checked for all 120 split objects.

The FSR pipeline generated ten repeated 80/20 holdouts using seeds 42–51. The interpolation partition was stratified by six target quantile bins when feasible and consistently contained 2,024 training and 507 test rows. For extrapolation, 50 candidate `GroupShuffleSplit` partitions were generated per repeat and the most target-balanced paper-disjoint candidate was retained. Each grouped training set contained 52 papers and each test set 14 papers, although row counts varied because paper sizes differed. Balance screening reduces avoidable changes in target distribution without weakening the defining paper-disjoint constraint. Integrity checks across all 20 partitions confirmed full coverage, nonempty partitions, no row overlap, and, for grouped partitions, no paper overlap.

3.6 Nested tuning and leakage control

Hyperparameters were selected separately inside every outer training partition. For ignition interpolation, inner stratified folds were used; grouped extrapolation and LOPO used stratified, paper-disjoint inner folds with both classes represented. For FSR, shuffled K-fold inner validation was used for interpolation and `GroupKFold` by paper for extrapolation. Up to 40 distinct parameter configurations were sampled from each family search space in the documented full workflow. Ignition used three inner grouped folds; FSR used five inner folds.

For ignition configuration h , the inner score ordered candidates by mean receiver-operating-characteristic area under the curve (ROC–AUC), then precision–recall AUC (PR–AUC), then lower Brier score. For FSR it ordered by lower root mean squared error (RMSE), then lower mean absolute error (MAE), then higher coefficient of determination R^2 . The best configuration was refitted on the complete outer training partition and evaluated once on the outer test partition. Search histories, selected hyperparameters, outer predictions, and integrity status were persisted. Dataset SHA-256 fingerprints linked the input, splits, and evaluations; the FSR evaluator explicitly aborted on a mismatch.

For the ignition task, decision thresholds were also selected without outer-test labels. Given a probability estimate $\hat{p}_i$,

$$\hat{y}_i(\tau) = \mathbb{I}(\hat{p}_i \geq \tau). \quad (3.2)$$

Four thresholds maximized, respectively, Matthews correlation coefficient (MCC), F1 score, balanced accuracy, and Youden’s $J = \text{sensitivity} + \text{specificity} - 1$ on the winning configuration’s inner out-of-fold predictions. Ties favored a threshold nearer 0.5. Once frozen, each threshold was applied to the outer test fold. No single threshold was declared universally optimal because each policy imposes a different balance between false positives and false negatives.

3.7 Metrics, uncertainty, and model selection

Threshold-independent classification metrics were ROC–AUC, PR–AUC, and the Brier score

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (3.3)$$

Thresholded metrics included MCC, F1, balanced accuracy, sensitivity, specificity, and precision. Regression metrics included

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (3.4)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{MBE} = \frac{1}{n} \sum_i (\hat{y}_i - y_i). \quad (3.5)$$

MAPE and normalized RMSE were also persisted. MAPE is unstable when measured FSR approaches zero and was therefore descriptive rather than a champion-selection criterion.

Reported “mean $\pm$ SD” values summarize outer-fold variability. In addition, 2,000-resample 95% bootstrap intervals were computed from persisted outer predictions. Repeated predictions for a row were averaged before resampling. Interpolation resampled unique rows; extrapolation resampled canonical papers, preserving the clustered inferential unit. Fold SD and cluster-bootstrap intervals answer different questions and are both reported where available.

Ignition candidates had to pass integrity checks, complete all required folds, produce valid probabilities, and have ROC–AUC fold SD no greater than 0.15. Interpolation was ranked on stratified-row ROC–AUC. Extrapolation was ranked on grouped ROC–AUC and additionally required successful LOPO evaluation. Values within 0.01 of the best were resolved by higher PR–AUC, lower uncertainty, and a declared simplicity order. FSR candidates required complete, finite evaluation, RMSE coefficient of variation no greater than 0.45, and mean $R^2 \geq -0.25$. Any RMSE within 2% of the best formed a tie band. Tie-breakers considered higher R^2 , lower uncertainty, and no augmentation; extrapolation additionally considered a smaller interpolation–extrapolation gap. Because all persisted FSR candidates used the same full feature set, the nominal physics-only and fewer-features tie-breakers did not distinguish them. For both tasks, refitting used the modal outer-fold hyperparameter configuration, with lexical resolution of an exact modal tie.

3.8 Post hoc explainability

Tree-model explanations were generated from persisted evaluation and refit artifacts, without retuning. Mean absolute SHAP magnitude was used as a global summary,

$$I_j^{\text{SHAP}} = \frac{1}{n} \sum_{i=1}^n |\phi_{ij}|, \quad (3.6)$$

where ϕ_{ij} is the additive contribution assigned to feature j for observation i . Permutation importance measured the degradation in held-out score after disrupting one original feature while leaving the fitted model fixed. For classification, positive values are losses in ROC–AUC; for regression, positive values are increases in RMSE. Repeated permutations provide the reported mean and standard deviation.

These measures are predictive, conditional on the fitted model and sampled data. They do not identify causal effects. Correlated physical and engineered variables can divide, substitute, or mask importance; one-hot levels and their parent categorical fields have different granularities; missingness can carry campaign information; and SHAP magnitude does not establish the sign of an effect. Negative permutation importance means that random disruption happened to improve held-out performance under the realized sample, not that the variable is physically protective. Explainability was consequently interpreted as evidence about model reliance and predictive associations only.

3.9 Reproducibility, estimands, and reporting conventions

Reproducibility was enforced at the level of data identity, partition membership, candidate definition, and randomization. The default random seed was 42; repeated splits used consecutive integer seeds. Split assignments stored the row and paper membership of every partition rather than relying on regeneration from an algorithm description. Candidate manifests recorded model family, weighting, fixed parameters, and augmentation condition. Search histories and selected hyperparameters preserved the choices made independently for each outer fold. Integrity files recorded failed fits or invalid outputs instead of replacing them with fabricated scores. This persisted-state design permits the report to be regenerated from evaluation outputs and fitted artifacts without rerunning or altering hyperparameter optimization.

The relevant estimand differs by protocol. In a row split, observations are the resampling units and papers can be represented in both partitions. Its performance is therefore conditional on the mixture of campaigns already present. In a grouped split, the held-out units are papers; rows within a paper remain dependent and should not be counted as independent evidence for cross-paper uncertainty. The use of canonical-paper bootstrap resampling for grouped predictions reflects this distinction. It does not eliminate all dependence—for example, multiple papers may share investigators, rigs, materials, or facilities—but it avoids the more serious error of assigning independent status to repeated rows from one publication.

Fold means, fold standard deviations, and pooled-prediction bootstrap estimates were kept conceptually separate. If m_k is a metric in outer fold k , the reported fold summary is

$$\bar{m} = \frac{1}{K} \sum_{k=1}^K m_k, \quad s_m = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (m_k - \bar{m})^2}. \quad (3.7)$$

This gives each repeat or fold one contribution. The bootstrap tables instead average repeated predictions for a row, recompute a metric on the pooled predictions, and resample the appropriate row or paper units. Because nonlinear metrics generally do not commute with averaging, a bootstrap point estimate need not reproduce $\bar{m}$.

The former describes variability among the frozen outer evaluations; the latter supplies a sampling interval under the stated cluster-resampling scheme.

All rates and errors were reported in the units of their persisted targets. Ignition discrimination metrics are dimensionless, and the Brier score is a mean squared probability error. FSR RMSE, MAE, and MBE retain m s^{-1} ; R^2 , MAPE, and NRMSE are dimensionless. Candidate names such as **baseline**, **sqrt**, and **inverse** identify computational recipes rather than physical hypotheses. Similarly, **rd_bt** denotes duplicated training observations sampled with replacement, not a second experimental cohort. These conventions prevent changes in evaluation protocol, aggregation, or units from being mistaken for changes in the underlying physical phenomenon.

4 Results

4.1 Data integrity and scope of inference

Both persisted evaluation suites passed their overall integrity checks. The ignition analysis used 4,487 deduplicated rows from 87 canonical papers, with 3,383 positive and 1,104 negative labels. Its frozen split manifest recorded no integrity failures across 120 split objects. The FSR analysis used 2,531 rows with numeric FSR from 66 papers, and all 20 repeated holdout partitions passed their coverage and separation checks. The corresponding SHA-256 fingerprints were `d80da0648ca3a7d77d31ae90ebea53b4e81b2ab28c8ec666ce6cd150ee7544f6` for ignition and `6e09a99e4ea69148e69151bd46149be4140d181b2b8294106927ea9499002fc8` for FSR. These fingerprints establish which persisted data versions the numerical results describe.

The principal result throughout this chapter is the difference between row-level interpolation and paper-grouped extrapolation. The former permits a publication to contribute observations to both sides of a split and is consequently a measure of prediction among represented campaigns. The latter excludes every test paper from training and is the relevant test of transfer to new literature campaigns. Neither protocol evaluates causal mechanisms, and neither extends the claim beyond the support of the compiled database.

4.2 Ignition classification

4.2.1 Model selection and interpolation performance

The selected interpolation model was `xgb_paper_sqrt`, an XGBoost classifier with paper weight $1/\sqrt{n_p}$ and no stated class-weight addition. Across the 15 repeated stratified outer folds, its ROC-AUC was 0.9076 ± 0.0102 , PR-AUC was 0.9687 ± 0.0039 , and mean Brier score was 0.1063. Its modal refit configuration used 800 trees, maximum depth 6, learning rate 0.05, column subsampling 0.6, row subsampling 1.0, minimum child weight 1, $\gamma = 0$, L_1 regularization 0.5, L_2 regularization 1.0, and the logistic objective. The supplementary three-repeat 80/20 holdout was consistent: ROC-AUC 0.9032 ± 0.0016 , PR-AUC 0.9663 ± 0.0015 , and Brier score 0.1114.

The leading XGBoost variants were tightly clustered in row interpolation (Table 4.1). The unweighted XGBoost baseline achieved 0.9058 ± 0.0091 ROC-AUC, only 0.0018 below the selected model. The class-weighted and combined square-root-weighted variants were similarly close. By contrast, the leading reported SVM reached 0.8608, the leading decision tree 0.8531, and the leading KNN 0.8325. The two reported focal-objective XGBoost recipes were near chance in ROC ranking (0.4960 and 0.4987), despite high PR-AUC values caused in part by the 75.4% positive prevalence. This combination illustrates why PR-AUC must be interpreted against prevalence and why candidate selection did not rely on it alone.

Table 4.1: Leading persisted ignition results. Values are outer-fold mean $\pm$ standard deviation; Brier is the fold mean.

Protocol	Candidate	ROC-AUC	PR-AUC	Brier
Row interpolation	<code>xgb_paper_sqrt</code>	0.9076 ± 0.0102	0.9687 ± 0.0039	0.1063
Row interpolation	<code>xgb_baseline</code>	0.9058 ± 0.0091	0.9682 ± 0.0036	0.1076
Row interpolation	<code>svm_weighted_sqrt</code>	0.8608 ± 0.0125	0.9518 ± 0.0049	0.1794
Row interpolation	<code>dt_baseline</code>	0.8531 ± 0.0192	0.9372 ± 0.0127	0.1296
Paper extrapolation	<code>xgb_baseline</code>	0.7183 ± 0.0444	0.8767 ± 0.0298	0.1791
Paper extrapolation	<code>xgb_class_only</code>	0.7036 ± 0.0447	0.8697 ± 0.0309	0.1873
Paper extrapolation	<code>xgb_paper_sqrt</code>	0.7004 ± 0.0365	0.8650 ± 0.0305	0.1866
Paper extrapolation	<code>svm_weighted_inverse</code>	0.6934 ± 0.0456	0.8787 ± 0.0218	0.1828

The row-resampled bootstrap gives a complementary uncertainty summary. After averaging repeated probabilities for each unique row, the interpolation champion had ROC-AUC 0.9122 (95% CI 0.9037–0.9209), PR-AUC 0.9703 (0.9660–0.9741), and Brier score 0.1023 (0.0969–0.1078). These bootstrap estimates need not equal the arithmetic means of the 15 fold metrics because they aggregate repeated predictions per row before computing each metric.

4.2.2 Paper-grouped extrapolation and the cross-paper gap

The extrapolation champion was the unweighted `xgb_baseline`. Its grouped outer-fold ROC-AUC was 0.7183 ± 0.0444 , PR-AUC 0.8767 ± 0.0298 , and Brier score 0.1791. The selected modal configuration was substantially more regularized and shallower than the interpolation champion: 400 trees, maximum depth 2, learning rate 0.01, column subsampling 0.6, row subsampling 0.6, minimum child weight 4, $\gamma = 0.1$, $L_1 = 0.5$, $L_2 = 1.0$, and the

logistic objective. The mandatory LOPO analysis was successful. Its paper-cluster bootstrap ROC–AUC was 0.7020 (95% CI 0.6376–0.7751), PR–AUC 0.8622 (0.8013–0.9222), and Brier score 0.1734 (0.1293–0.2112).

For the primary grouped protocol, the same paper-cluster procedure yielded ROC–AUC 0.7101 (95% CI 0.6408–0.7848), PR–AUC 0.8683 (0.8110–0.9277), and Brier score 0.1713 (0.1234–0.2124). These intervals are considerably wider than the interpolation intervals, reflecting both fewer independent paper units and heterogeneity among campaigns.

The unweighted baseline’s ROC–AUC fell from 0.9058 under stratified row interpolation to 0.7183 under grouped extrapolation, an absolute reduction of 0.1875. For the interpolation champion itself, the corresponding reduction was larger: 0.9076 to 0.7004, or 0.2072. Combined class and paper weighting did not remove the gap; the square-root-weighted XGBoost model dropped by 0.2215. The decision-tree baseline had the largest reported reduction, 0.2414. Linear SVM variants showed smaller numerical drops of 0.0746–0.0855, but this was because their interpolation scores were already lower, not because their grouped scores were superior. The focal variants’ near-zero gaps likewise reflect near-chance performance in both settings and are not evidence of robust transfer.

Table 4.2: Ignition cross-paper generalization gap. Positive values denote lower ROC–AUC when whole papers were held out.

Candidate	Row ROC–AUC	Grouped ROC–AUC	Absolute drop
xgb_baseline	0.9058	0.7183	0.1875
xgb_class_only	0.9055	0.7036	0.2019
xgb_paper_sqrt	0.9076	0.7004	0.2072
xgb_paper_inverse	0.9008	0.6934	0.2074
xgb_weighted_sqrt	0.9057	0.6842	0.2215
dt_baseline	0.8531	0.6117	0.2414

This deterioration is the clearest classification result. Rows within represented papers can be ranked with high accuracy, but transfer to unseen papers is only moderate and more uncertain. The change in selected candidate reinforces the point: square-root paper weighting narrowly led within-paper interpolation, whereas the plain baseline generalized best under the declared grouped policy. A single “best ignition model” without its evaluation protocol would therefore be misleading.

4.2.3 Operational thresholds

Threshold choice produced materially different operating characteristics (Table 4.3). For the interpolation champion, the mean MCC threshold was 0.7920, giving sensitivity 0.7916, specificity 0.8723, and precision 0.9500 over persisted outer predictions. The F1-optimal threshold was much lower at 0.3688; sensitivity increased to 0.9516 while specificity fell to 0.4798. Balanced accuracy and Youden’s J selected the same mean threshold, 0.8379, with sensitivity 0.7660 and specificity 0.9013.

For the extrapolation champion, the MCC threshold of 0.8099 produced sensitivity 0.7290 and specificity 0.5836. The F1 threshold, 0.2850, classified nearly every row as positive: sensitivity was 0.9837 but specificity only 0.0592. Balanced accuracy and Youden’s J again coincided, at threshold 0.8583, balancing sensitivity 0.6639 and specificity 0.6621. This sensitivity–specificity exchange is especially consequential in an imbalanced data set. The persisted results support retaining all four policies for application-specific decisions rather than presenting one threshold as universally preferable.

Table 4.3: Thresholded performance of the selected ignition classifiers. Counts pool repeated outer-test predictions and therefore exceed the number of unique database rows.

Champion	Policy	$\bar{\tau}$	Sens.	Spec.	Precision	FP	FN
Interpolation	MCC	0.7920	0.7916	0.8723	0.9500	423	2,115
Interpolation	F1	0.3688	0.9516	0.4798	0.8486	1,723	491
Interpolation	Balanced accuracy	0.8379	0.7660	0.9013	0.9596	327	2,375
Extrapolation	MCC	0.8099	0.7290	0.5836	0.8429	1,379	2,750
Extrapolation	F1	0.2850	0.9837	0.0592	0.7621	3,116	165
Extrapolation	Balanced accuracy	0.8583	0.6639	0.6621	0.8576	1,119	3,411

4.2.4 Ignition model reliance

SHAP and permutation rankings presented complementary descriptions of the XGBoost models. In the interpolation champion, the largest mean absolute SHAP magnitudes were oxygen (0.6150), Prandtl number (0.4154), absolute flow speed (0.3429), gas molar mass (0.3140), ignition time (0.2521), fuel pyrolysis temperature (0.2241), and

the third sample dimension (0.2140). In the extrapolation champion, oxygen remained first (0.3591), followed by absolute flow speed (0.2500), Prandtl number (0.2497), gas molar mass (0.2047), fuel pyrolysis temperature (0.2044), ignition time (0.1638), and signed flow velocity (0.1271). These are average absolute contribution magnitudes in the fitted score space; they do not indicate whether larger values increase or decrease ignition probability.

Held-out permutation importance gave a different ordering. For the extrapolation champion, whose unpermuted held-out ROC-AUC in this analysis was 0.7818, the largest mean losses were obtained for fuel pyrolysis temperature 0.0977 ± 0.0051 , ignition time 0.0452 ± 0.0066 , fuel specific heat 0.0279 ± 0.0019 , thermal diffusion time 0.0206 ± 0.0020 , the third sample dimension 0.0158 ± 0.0021 , and absolute flow speed 0.0130 ± 0.0085 . Oxygen, despite leading SHAP magnitude, had permutation importance -0.0054 ± 0.0085 . Likewise, Prandtl number was -0.0200 ± 0.0067 . This disagreement is expected when predictors are correlated or substitutable: a model can allocate substantial local contributions to a variable even if shuffling that variable does not degrade aggregate ranking because related variables preserve information. It would be incorrect to translate either ranking into a causal hierarchy of combustion controls.

4.3 Flame-spread-rate regression

4.3.1 Interpolation model comparison

The declared selection policy chose the unaugmented random-forest baseline `rf_baseline` for row interpolation. Across ten repeated row holdouts it achieved RMSE $0.006263 \pm 0.000472 \text{ m s}^{-1}$, MAE $0.002103 \pm 0.000173 \text{ m s}^{-1}$, and $R^2 = 0.8790 \pm 0.0178$. Its mean bias error was $-3.88 \times 10^{-5} \text{ m s}^{-1}$. The modal configuration used 500 trees, maximum depth 10, 0.8 of the available features at each split, minimum leaf size 1, and minimum split size 2.

The numerically lowest interpolation RMSE belonged to bootstrap-augmented `xgb_baseline`, $0.006236 \pm 0.000698 \text{ m s}^{-1}$, but it lay within the 2% tie band and had $R^2 = 0.8783 \pm 0.0310$. The unaugmented random forest was selected after the higher- R^2 , lower-uncertainty, and no-augmentation tie-breakers. Other competitive results included unaugmented `xgb_absoluteerror_sqrt`, with RMSE $0.006338 \pm 0.000680 \text{ m s}^{-1}$, MAE $0.002028 \pm 0.000166 \text{ m s}^{-1}$, and $R^2 = 0.8752 \pm 0.0294$. KNN, MLP, and SVR candidates were materially weaker in the leading persisted rankings: the best listed KNN condition had RMSE 0.0090, the best listed MLP 0.0102, and the best listed SVR 0.0106 m s^{-1} .

The row bootstrap for the selected random forest, after averaging repeated predictions per row, estimated RMSE $0.005955 \text{ m s}^{-1}$ (95% CI 0.004734–0.007146), MAE $0.001973 \text{ m s}^{-1}$ (0.001759–0.002215), and $R^2 = 0.8845$ (0.8449–0.9198). As for classification, these pooled-prediction estimates differ slightly from the mean of foldwise metrics because the order of aggregation and metric calculation is different.

4.3.2 Unseen-paper performance and uncertainty

The same unaugmented random-forest candidate was selected for paper extrapolation, but with a distinct modal configuration: 500 trees, maximum depth 12, feature fraction 0.5, minimum leaf size 8, and minimum split size 2. Its ten grouped holdouts yielded RMSE $0.012205 \pm 0.004115 \text{ m s}^{-1}$, MAE $0.004819 \pm 0.001144 \text{ m s}^{-1}$, and $R^2 = 0.4576 \pm 0.2215$. Mean bias was $-0.000658 \text{ m s}^{-1}$. Fold-level ranges emphasize instability across paper composition: RMSE varied from 0.007775 to 0.022121 m s^{-1} , while R^2 ranged from 0.0176 to 0.7465.

Paper-cluster bootstrap intervals were correspondingly broad. The grouped random-forest estimate was RMSE $0.011867 \text{ m s}^{-1}$ (95% CI 0.006525–0.017839), MAE $0.004482 \text{ m s}^{-1}$ (0.002889–0.006528), and $R^2 = 0.4893$ (0.2773–0.7319). The interval width is part of the scientific result: transfer accuracy depends strongly on which papers are withheld, and the effective extrapolation sample comprises papers, not thousands of exchangeable rows.

The nearest grouped competitors did not clearly overturn the random-forest selection. Augmented `rf_baseline` had RMSE $0.012218 \pm 0.004060 \text{ m s}^{-1}$ and $R^2 = 0.4477 \pm 0.2468$. Augmented `xgb_paper_sqrt` had RMSE $0.012572 \pm 0.003832 \text{ m s}^{-1}$ and $R^2 = 0.4411 \pm 0.1548$. Augmented `xgb_absoluteerror_inverse` had RMSE $0.012632 \pm 0.003872 \text{ m s}^{-1}$ and $R^2 = 0.4223 \pm 0.1973$. Six candidate-condition combinations were rejected by the extrapolation policy: both data conditions for `mlp_log_target_sqrt`, `mlp_log_target_inverse`, and `svr_linear`. Their mean grouped R^2 values ranged from -0.696 to -3.093 , and several also exceeded the RMSE coefficient-of-variation limit.

Table 4.4: Leading persisted FSR results in original target units. Values are outer-fold mean $\pm$ standard deviation.

Protocol	Candidate-condition	RMSE (m s^{-1})	MAE (m s^{-1})	R^2
Row interpolation	<code>rf_baseline-rd</code>	0.006263 ± 0.000472	0.002103 ± 0.000173	0.8790 ± 0.0178
Row interpolation	<code>xgb_baseline-rd_bt</code>	0.006236 ± 0.000698	0.002226 ± 0.000367	0.8783 ± 0.0310
Row interpolation	<code>xgb_absoluteerror_sqrt-rd</code>	0.006338 ± 0.000680	0.002028 ± 0.000166	0.8752 ± 0.0294
Paper extrapolation	<code>rf_baseline-rd</code>	0.012205 ± 0.004115	0.004819 ± 0.001144	0.4576 ± 0.2215
Paper extrapolation	<code>rf_baseline-rd_bt</code>	0.012218 ± 0.004060	0.004856 ± 0.001266	0.4477 ± 0.2468
Paper extrapolation	<code>xgb_paper_sqrt-rd_bt</code>	0.012572 ± 0.003832	0.005130 ± 0.000882	0.4411 ± 0.1548

4.3.3 FSR cross-paper gap and augmentation

For the selected random forest, grouped RMSE exceeded row-interpolation RMSE by $0.005942 \pm 0.004138 \text{ ms}^{-1}$, almost doubling from 0.006263 to 0.012205 ms^{-1} . Its R^2 fell from 0.8790 to 0.4576 , a reduction of 0.4214 . MAE similarly increased from 0.002103 to 0.004819 ms^{-1} . The gap is not attributable to a change of metric or model family: it appears for the same candidate evaluated under the two split mechanisms. It therefore quantifies the optimism that would result from describing random-row scores as new-paper performance.

Bootstrap augmentation did not provide a systematic advantage. For the selected random forest, augmentation changed interpolation RMSE by $+0.0000265 \pm 0.0001985 \text{ ms}^{-1}$ and grouped RMSE by $+0.0000131 \pm 0.0004498 \text{ ms}^{-1}$, where positive values favor the unaugmented condition. Some recipes improved: grouped `xgb_paper_sqrt` changed by $-0.000392 \pm 0.000690 \text{ ms}^{-1}$, and `xgb_log_target_inverse` by $-0.000665 \pm 0.001507 \text{ ms}^{-1}$. Others worsened, including grouped `xgb_baseline` by $+0.000554 \pm 0.000615 \text{ ms}^{-1}$. The magnitudes were generally small relative to between-fold uncertainty. The selection of `rd` for both champions is thus consistent with the empirical results and the policy preference for real data within the tie band. It does not imply that bootstrap copies were invalid; it shows that resampling observed training rows did not robustly solve transfer to unseen papers.

4.3.4 Paper-level heterogeneity

Aggregated performance concealed severe paper-specific errors. For the grouped random-forest champion, the hardest reported paper was `doi::10.1016/j.combustflame.2020.12.005`, with RMSE 0.13313 ms^{-1} , MAE 0.12536 ms^{-1} , and $R^2 = -13.35$. The next paper had RMSE 0.07729 ms^{-1} . Several others had RMSE near 0.02 ms^{-1} , while many papers were predicted with errors below 0.003 ms^{-1} . Extremely negative per-paper R^2 values also occurred for papers with little within-paper target variance; in that setting, a modest absolute error can be large relative to the paper-specific variance denominator. Accordingly, per-paper RMSE and MAE are more stable indicators of absolute predictive error than isolated per-paper R^2 . The heterogeneous error distribution is consistent with the wide grouped fold SD and bootstrap intervals and argues against a universal accuracy claim.

4.3.5 FSR model reliance

The random-forest champion’s impurity-based importance placed flow velocity first (0.2187), followed by thermal diffusion time (0.1830), oxygen concentration (0.1013), characteristic length (0.0830), the parsed sample-dimension field (0.0698), gas thermal conductivity (0.0658), Péclet number (0.0476), and Reynolds number (0.0402). Impurity importance is included as a model summary but is known to favor variables that offer many potential split points and does not measure held-out dependence.

Mean absolute SHAP magnitude gave thermal diffusion time the largest value, 0.005293 ms^{-1} , followed by oxygen concentration 0.003105 , gas thermal conductivity 0.001441 , flow velocity 0.001089 , fuel volumetric heat capacity 0.001034 , fuel density 0.000949 , core volumetric heat capacity 0.000852 , and core density 0.000787 ms^{-1} . These magnitudes summarize how strongly features contributed to predictions in the evaluated artifact; taking absolute values removes direction.

Permutation importance was computed on a held-out set whose unpermuted RMSE was 0.010919 ms^{-1} . The clearest positive degradations were oxygen concentration, $0.002276 \pm 0.000349 \text{ ms}^{-1}$; flow velocity, 0.001486 ± 0.000782 ; gas thermal conductivity, 0.001137 ± 0.000083 ; and thermal diffusion time, $0.000967 \pm 0.000682 \text{ ms}^{-1}$. Prandtl number, gas specific heat, and Reynolds number had smaller positive values of 0.000271 , 0.000156 , and 0.000154 ms^{-1} , respectively. Several correlated thermophysical variables had negative permutation values, including fuel volumetric heat capacity (-0.000837), fuel density (-0.000790), and core volumetric heat capacity ($-0.000665 \text{ ms}^{-1}$). These negative values indicate that permutation happened to reduce RMSE in this held-out sample; they do not establish an inverse physical effect.

Across the three summaries, oxygen concentration, flow, gas conductivity, and thermal diffusion time repeatedly appeared near the top, which is evidence that the fitted predictor relied on information associated with those variables. Nevertheless, the database contains strongly related raw and derived transport quantities, campaign-specific missingness, and uneven coverage of materials and conditions. Importance can therefore migrate among correlated predictors, and none of the rankings identifies a causal intervention. In particular, the results do not demonstrate that changing one variable while holding all else fixed would produce the corresponding change in FSR.

4.4 Synthesis

The two tasks exhibit the same structural result despite different targets, cohorts, models, and metrics. Ignition ROC-AUC declined from approximately 0.91 under row interpolation to 0.72 for the grouped champion. FSR RMSE approximately doubled and R^2 fell from 0.879 to 0.458 when complete papers were withheld. Uncertainty also expanded under grouped resampling. These gaps are direct evidence that campaign membership captures dependencies not eliminated by the standardized predictor set.

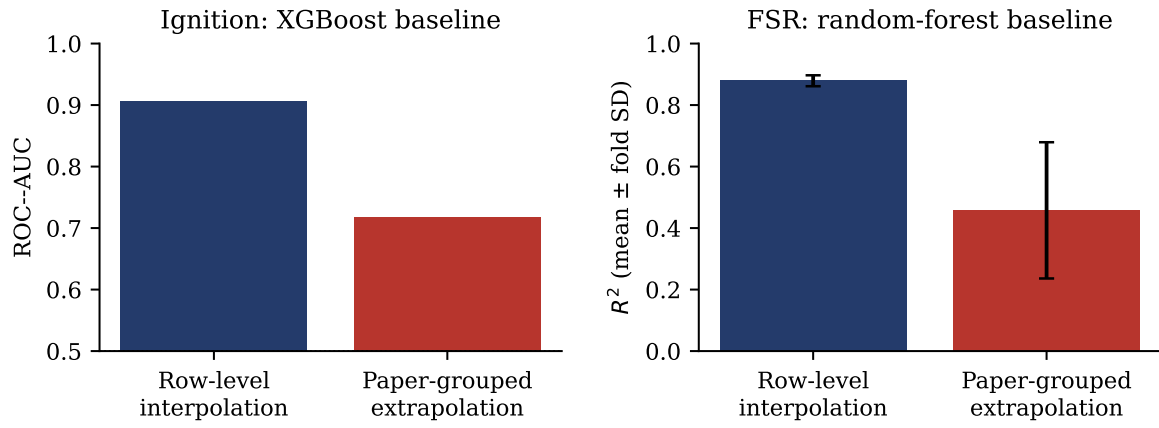


Figure 4.1: Protocol sensitivity for the common baseline configurations used to isolate the row-to-paper shift. Error bars in the FSR panel are fold standard deviations. The ignition panel compares the same unweighted XGBoost candidate under both protocols. Values are generated from persisted report tables.

The selected families were tree ensembles in all cases, but the configurations and weighting choices differed by prediction question. Paper weighting narrowly won ignition interpolation yet not ignition extrapolation. The random-forest baseline won both FSR protocols, with a deeper but much larger minimum leaf size for grouped transfer. Bootstrap augmentation changed some regression candidates in either direction but produced no general reduction in cross-paper error. Threshold optimization substantially changed ignition sensitivity and specificity, so deployment requires an explicit operating objective.

These findings support a bounded conclusion. The database contains enough repeated structure for strong within-campaign prediction and enough transferable signal for performance above naive baselines on average across unseen papers. However, the cross-paper penalties, wide grouped uncertainty, and extreme paper-level failures preclude universal claims. SHAP and permutation analyses identify predictive reliance, not causal combustion laws. The defensible outputs are therefore protocol-specific estimates with their uncertainty and documented feature context, rather than a single headline score or a mechanistic claim.

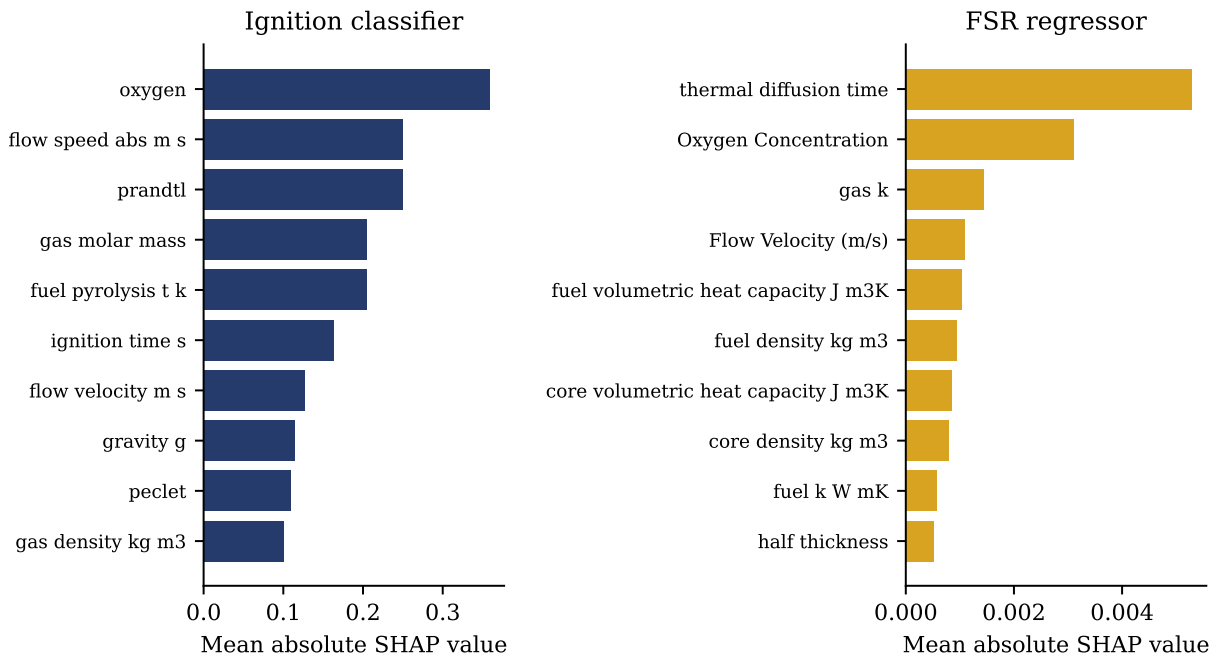


Figure 4.2: Top ten global mean absolute SHAP magnitudes in the persisted explainability tables. Magnitudes are task- and model-specific and are not comparable across panels. Absolute SHAP values quantify predictive contribution without direction and do not establish causal influence.

4.5 Robustness of comparative conclusions

The principal comparisons were not artifacts of one secondary metric. For ignition, the extrapolation champion had the highest grouped ROC–AUC, the lowest grouped Brier score among the three leading XGBoost rows shown in Table 4.1, and a successful LOPO result. The `svm_weighted_inverse` candidate had a slightly higher grouped PR–AUC (0.8787 versus 0.8767) but lower ROC–AUC (0.6934 versus 0.7183). This does not contradict selection because PR–AUC was a tie-breaker rather than the primary metric, and the SVM was outside the 0.01 ROC–AUC tie tolerance. Likewise, `xgb_paper_sqrt` had the smallest fold SD among the three leading XGBoost candidates (0.0365), but its lower grouped mean prevented uncertainty alone from making it champion.

Calibration-oriented evidence also deteriorated across papers. The selected interpolation classifier’s mean Brier score was 0.1063, whereas the extrapolation champion’s grouped mean was 0.1791. Because the selected candidates differ, this is not a pure paired model effect, but it agrees with the discrimination gap: probability estimates were less accurate when campaigns were excluded. The threshold results add a practical warning. A policy optimized for F1 under high positive prevalence retained excellent sensitivity yet accepted almost no specificity in grouped evaluation. Reporting only F1 or sensitivity would conceal that behavior.

For FSR, RMSE and MAE told the same broad story. The selected model’s grouped RMSE was 1.95 times its interpolation RMSE, and grouped MAE was 2.29 times its interpolation MAE. The foldwise mean R^2 reduction was 0.4214. Mean MBE remained small in absolute terms under interpolation but became more negative in grouped evaluation, indicating average underprediction of approximately $6.6 \times 10^{-4} \text{ m s}^{-1}$ over the grouped folds. Small global bias did not imply small individual error: grouped MAE was more than seven times the magnitude of MBE because positive and negative residuals cancel in the latter.

MAPE was not used to overturn these conclusions. The interpolation random-forest MAPE was already 73.8%, and its grouped MAPE was 357.0%, despite comparatively small absolute errors. This occurs because the FSR cohort includes zero and near-zero values, for which percentage denominators are undefined or unstable. NRMSE similarly rose from 0.697 to 1.433 between protocols, but its scale depends on the normalization convention. RMSE in m s^{-1} , MAE in the same units, and R^2 were therefore the most interpretable joint summary.

Finally, the selected FSR point estimates were not uniquely separated from every competitor. Several tree ensembles occupied a narrow RMSE range, especially relative to the grouped fold SD of roughly 0.004 m s^{-1} . Champion status should be understood as the deterministic outcome of the predeclared tie policy, not proof that one fitted algorithm is intrinsically superior in all future samples. The stable conclusion is broader: ensemble trees formed the leading group in these persisted evaluations, while the protocol shift from rows to papers was much larger than the differences among the leading candidate recipes.

5 Discussion

The central result of this study is not that one algorithm “solves” material fire safety, but that the apparent predictive difficulty depends strongly on what is held out. When rows were divided at random, experiments from the same publication could occur in both training and test sets. Under that interpolation question, the flame-spread-rate (FSR) random-forest champion reached $R^2 = 0.879 \pm 0.018$ with an RMSE of $0.00626 \pm 0.00047 \text{ m s}^{-1}$, while the ignition XGBoost champion reached a ROC-AUC of 0.908 ± 0.010 . When complete papers were held out, the corresponding results fell to $R^2 = 0.458 \pm 0.222$ and RMSE $0.01221 \pm 0.00412 \text{ m s}^{-1}$ for FSR, and ROC-AUC 0.718 ± 0.044 for ignition. The FSR error approximately doubled, and the ignition ROC-AUC decreased by about 0.19. This consistent row-versus-paper generalization collapse is the most consequential scientific finding. It indicates that models can interpolate effectively among conditions resembling a campaign already represented in training, but only partially transfer to an unseen publication, apparatus, laboratory practice, or sampled regime.

This chapter interprets that finding under four deliberately visible allocations: limitations (approximately 30%), comparison with the literature (approximately 30%), real-world applications (approximately 30%), and future work (approximately 10%). These proportions reflect the practical importance of defining what the models do not establish, placing their scores beside protocol-compatible evidence, and translating them into a cautious decision-support role.

5.1 Limitations (approximately 30%)

5.1.1 Database heterogeneity and the unit of generalization

The source database is a synthesis of published experiments rather than a prospectively designed factorial study. The FSR analysis contains 2,531 usable target rows from 66 canonical papers; the ignition analysis contains 4,487 deduplicated rows from 87 papers, of which 3,383 report ignition and 1,104 report no ignition. These counts provide considerably more observations than a single campaign, but rows are not independent replications of a common data-generating process. A publication may contribute many nearby settings from one rig, material batch, diagnostic method, gravity platform, or laboratory. Another may contribute only a handful of observations in a distinct regime. Consequently, a row-level split can place near-neighbours—and campaign-specific signatures—on both sides of the evaluation boundary.

The grouped protocol addresses the clearest version of this dependence by constructing canonical paper identities from DOI or citation and prohibiting paper overlap. It therefore tests a harder and more relevant question: transfer to a publication that was absent from model fitting and inner hyperparameter selection. It does not, however, guarantee transfer to every future spacecraft material or operating envelope. Papers can share laboratories, investigators, facilities, wire stocks, reporting conventions, or inherited correlations. Conversely, two campaigns in one paper may differ materially. “Paper” is thus a defensible, reproducible proxy for experimental source, not a perfect ontological definition of domain. A still stronger validation would hold out laboratories, facilities, material families, gravity platforms, or time periods, but the current metadata do not support all such partitions consistently.

The collapse under paper holdout is evidence that publication identity carries predictive structure beyond the intended physical variables. That structure may include legitimate regime information omitted from the schema, such as apparatus heat losses or material morphology, as well as source-specific measurement and reporting practices. The result should not be described simply as overfitting in the narrow algorithmic sense. It is more accurately a combination of source shift, incomplete covariate description, clustered sampling, and possible outcome-definition differences. The large FSR fold variability— R^2 ranging from approximately 0.018 to 0.746 for the grouped champion—further shows that transfer difficulty depends on which papers are withheld.

5.1.2 Missingness, measurement comparability, and representation

Literature synthesis inevitably produces sparse and uneven feature coverage. Material thermophysical properties may be tabulated, inferred from material labels, or absent. Geometry can be reported in formats that are not directly commensurate across wires, flat samples, cylinders, and spheres. Pressure, flow, oxygen concentration, ignition settings, and gravity are not varied independently, and some are fixed or omitted within individual campaigns. The preprocessing contract handles this computationally: numeric values are median-imputed for most families, categorical missingness is assigned an “Unknown” level, and XGBoost can use native numeric missingness. Computational handling is not equivalent to recovering the unobserved physics.

Missingness may itself be source-dependent. If one laboratory systematically reports a property and another does not, a tree can use the missingness pattern as a source fingerprint. This can improve random-split prediction while harming transfer. Median imputation also compresses genuinely different unknown cases to a common value, while a constant categorical level conflates “not measured”, “not applicable”, and “not reported”. The grouped results therefore integrate both the difficulty of physical extrapolation and the consequences of imperfect harmonisation.

Feature engineering improves consistency but does not remove these concerns. The ignition pipeline uses 32 numeric and five categorical features, including material, gas-transport, operating-condition, geometry, flow-direction, gravity-regime, and ignition-method descriptors. The FSR pipeline detects the available numeric and categorical roles after excluding explicit source identifiers, duplicate target fields, and post-outcome variables. Derived quantities such as Reynolds, Peclet, Prandtl, volumetric heat capacity, thermal diffusivity, and thermal diffusion time can encode useful physical structure. Yet derived features share inputs and can be mutually dependent; their values are only as reliable as the source measurements and assumptions used to calculate them.

One schema discrepancy requires explicit qualification: the persisted FSR feature manifest includes categorical **Ignition**, although the pipeline documentation describes ignition fields as excluded. Its reported tree and SHAP importance is zero, but that does not substitute for a controlled ablation. Future FSR evaluation should remove the field and regenerate frozen comparisons before asserting that the documented and executed schemas are identical.

5.1.3 Imbalance and observational sampling

Two forms of imbalance matter. First, ignition is a majority-positive task: approximately three quarters of the 4,487 records are ignition cases. A high PR-AUC is therefore partly facilitated by prevalence and must be read beside ROC-AUC, Brier score, sensitivity, specificity, and threshold-dependent performance. The four stored threshold policies make the trade-off explicit. For example, the grouped champion’s F1-oriented policy attains very high aggregate sensitivity but very low specificity, whereas the balanced-accuracy policy gives much more even sensitivity and specificity. No threshold is universally preferable; its acceptability depends on the consequences of false negatives and false positives.

Second, papers contribute unequal row counts. A densely sampled campaign can dominate empirical risk even if it represents a narrow region. The candidate portfolios test no, square-root, and inverse paper weights, and weight-blind families use deterministic probability-proportional resampling. These mechanisms change each source’s influence, but cannot manufacture missing regimes. The selected grouped champions were the unweighted baselines: random forest for FSR and XGBoost for ignition. This does not prove that source imbalance is unimportant. It only shows that the tested weighting corrections did not improve the selection objective enough under these frozen splits.

Observational coverage creates a more fundamental limitation. Oxygen, pressure, gravity, flow, geometry, material properties, and ignition practice were chosen by investigators for particular scientific questions. Their joint distribution is therefore neither random nor uniform. For example, a material may occur mainly in one geometry and gravity regime, while a pressure range may be associated with one facility. A flexible learner can predict from these correlations without separating their physical effects. Sparse partial-gravity observations are especially problematic because lunar and Martian gravity conditions cannot be treated as densely sampled interpolation domains merely because “gravity” is a numeric feature.

5.1.4 Model dependence, metrics, and uncertainty

Model rankings are contingent on the tested candidates, search spaces, preprocessing, loss functions, seeds, and selection policy. The configuration files declare 34 FSR and 26 ignition candidates, but the persisted merged FSR manifest contains 25 candidates and the formal ignition selection record omits the available MLP shards. The selected FSR champion was random forest rather than XGBoost, while ignition selected XGBoost. These results apply to the persisted comparison pools; they neither establish complete execution of every configured recipe nor prove that the selected model is optimal among all possible algorithms.

Likewise, a point estimate is not a complete account of performance. R^2 is sensitive to the target variance in a held-out fold and can be unstable when a small set of papers defines a narrow or shifted range. FSR MAPE is extremely large because percentage error is ill-conditioned when observed spread approaches zero; RMSE and MAE in physical units are more interpretable here. For ignition, ROC-AUC measures ranking across thresholds but not probability calibration or a mission-specific operating point. The paper-cluster bootstrap, repeated splits, per-paper metrics, calibration outputs, and fold standard deviations should therefore accompany any headline number.

The final refitted artifacts also have a different evidential status from the outer-fold results. Refitting uses all labelled rows and is appropriate for producing a deployable model, but its training fit is not an unbiased estimate of future performance. Only the nested, untouched outer-fold predictions support comparative claims. The model cards correctly preserve this distinction.

5.1.5 SHAP and the boundary between explanation and causation

The explainability outputs describe model behaviour, not causal mechanisms. For ignition, the global XGBoost SHAP ranking assigns high mean absolute contributions to oxygen, absolute flow speed, Prandtl number, gas molar mass, fuel pyrolysis temperature, and ignition time. Yet grouped permutation importance places fuel pyrolysis temperature, ignition time, fuel heat capacity, thermal diffusion time, and a sample dimension at the top. For FSR, SHAP ranks thermal diffusion time and oxygen prominently, while held-out permutation importance ranks oxygen, flow velocity, gas thermal conductivity, and thermal diffusion time most strongly. Differences are expected because SHAP and permutation importance answer different questions and react differently to correlated inputs.

A SHAP value is an additive decomposition of a particular fitted function relative to a background distribution. It does not show that intervening on that feature would produce the attributed change. Correlated and derived variables allow credit to be redistributed among substitutes. Source confounding can make an apparatus-correlated variable appear physically decisive. Missingness patterns may attract attribution. Mean absolute SHAP also suppresses direction, interactions, and regime dependence. The physically plausible appearance of many rankings is reassuring, but plausibility is not validation of a causal law. Claims such as “oxygen causes the predicted increase” require intervention, mechanistic identification, or a causal design that this observational database does not provide.

The safest interpretation is predictive: within the fitted model and sampled distribution, these features are influential for separating or estimating outcomes. The permutation results add a transfer-oriented check by measuring score degradation on held-out data, but they remain model- and sample-dependent. Explainability is therefore best used to audit whether the learner relies on obviously inappropriate signals, to generate hypotheses, and to identify data-collection priorities. It should not replace combustion theory or controlled experimentation.

5.2 Comparison with the Literature (approximately 30%)

5.2.1 Rivera/Jose wire-specific machine learning

Rivera, Jose, and colleagues report a literature-derived electrical-wire FSR study in which categorical regime descriptors and augmentation improve random ten-fold cross-validation performance, with several nonlinear models exceeding $R^2 = 0.9$ [4]. Their database contains approximately 1,200 wire observations and explicitly represents core and insulation materials, spread direction, and configuration. Categorisation raises the reported performance of gradient boosting, KNN, SVM, decision trees, and MLP, while bootstrap augmentation is particularly effective for gradient boosting and KNN. This supports two conclusions also visible here: flame spread is strongly nonlinear, and categorical information can expose regime structure that numeric properties alone may not capture.

The numerical results must not be presented as if they were generated by equivalent protocols. The Rivera/Jose work uses random cross-validation and explicitly characterises its result as robustness within the compiled database rather than strict leave-source-out transfer. Its scope is electrical wires, and its target is reported in mm s^{-1} . The present FSR database is broader in geometry and source coverage, reports the model target in m s^{-1} , and makes paper-disjoint nested evaluation the primary scientific test. Hyperparameter search, feature definitions, data versions, augmentation, and source composition also differ. A direct statement that 0.458 is “worse than” > 0.9 would confound task and protocol.

The fair comparison is question-specific. On row-level interpolation, this study’s random-forest $R^2 = 0.879 \pm 0.018$ is close to, though generally below, the strongest wire-specific random-CV values. On paper-level extrapolation, the same family falls to $R^2 = 0.458 \pm 0.222$. Rivera/Jose do not report an equivalent paper-disjoint score, so their > 0.9 result neither contradicts nor establishes performance on this harder transfer task. Instead, their stated caveat and the present protocol gap jointly reinforce the need to name the validation unit whenever quoting accuracy.

There is also a substantive difference in what categorisation can encode. Wire-specific labels such as insulation chemistry, core material, concurrent/opposed spread, and orientation can partition physically distinct regimes. In a broader database, adding similarly detailed categories may improve prediction, but it may also increase sparsity and source fingerprinting. The present models exclude explicit paper identifiers and post-outcome features, yet legitimate categories can remain highly campaign-specific. Their value should therefore be tested under grouped validation, not assumed from random-CV gains.

The augmentation comparison is similarly instructive. Rivera/Jose find bootstrap resampling particularly useful, while noise injection and GAN augmentation can create statistically plausible but not necessarily physically consistent combinations. The present FSR benchmark treats bootstrap copies as reweighting observed conditions; copies inherit the source paper and are never added to the test set. Its selected FSR champions use the real-data condition rather than bootstrap augmentation. Thus, augmentation did not compensate for source shift in the selected model. This supports a conservative principle: resampling can alter optimisation and density, but it cannot supply a new material, apparatus, or gravity regime.

5.2.2 XGBoost and the tabular-learning literature

The strength of tree ensembles is consistent with broader tabular-learning evidence. Shwartz-Ziv and Armon compare XGBoost with several deep tabular architectures on eleven datasets and find that XGBoost generally outperforms individual deep models, requires less tuning, and is less sensitive to whether a dataset appeared in the proposing model’s original paper [5]. The present database has precisely the characteristics that often favour tree ensembles: mixed numeric and categorical predictors, nonlinear interactions, sparse cells, missing fields, unequal scales, and modest sample size relative to the variety of regimes.

The ignition outcome fits this expectation strongly. XGBoost occupies the top of both interpolation and grouped leaderboards; the paper-square-root variant is selected for interpolation and the unweighted baseline for extrapolation. By comparison, decision tree, KNN, and SVM candidates generally rank lower, while the tested focal XGBoost variants perform near chance. This last result is a reminder that a generally strong architecture does not rescue an unsuitable objective implementation or weighting combination.

The FSR result is more nuanced. Random forest narrowly leads interpolation and is selected for grouped transfer, while several XGBoost variants remain competitive. The selected random forest uses no paper weighting, no augmentation, and standard identity-target regression. Its grouped R^2 is 0.458, compared with 0.441 for the strongest listed augmented paper-weighted XGBoost candidate. This difference is small relative to fold variation and should not be elevated into a universal claim that bagging is superior to boosting. It shows only that, within the tested portfolio and policy, the random-forest baseline offered the preferred accuracy–stability combination.

The MLP candidates do not outperform the tree ensembles, particularly under grouped FSR evaluation, where their mean R^2 values can become negative and highly variable. This agrees directionally with the tabular literature, but the explanation is not that neural networks are intrinsically inappropriate. Data volume, regularisation, architecture, optimisation, and representation all matter. Shwartz-Ziv and Armon also find that ensembles combining deep models with XGBoost can outperform either component alone [5]. The present work did not evaluate a formally stacked, outer-fold-safe heterogeneous ensemble, so it cannot rule out such gains.

5.2.3 Physics-informed learning and combustion knowledge

Physics-informed neural networks constrain learning with residuals of governing differential equations and boundary or initial conditions [3]. More broadly, physics-informed machine learning seeks to combine data with conservation laws, symmetries, constitutive knowledge, or admissibility constraints [2]. This is attractive for flame spread because experimental coverage is sparse, predictions must remain physically plausible, and extrapolation is precisely where unconstrained correlations are weakest.

The present models are physics-featured rather than physics-constrained. Inputs include thermophysical properties, dimensionless groups, geometry, oxygen, pressure, flow, gravity, and ignition conditions, but the learned function is not required to satisfy mass, species, momentum, or energy balances. Optional monotonic constraints on oxygen exist for selected XGBoost configurations, yet a monotonic feature constraint is not equivalent to a combustion model. Likewise, categorising gravity or spread direction supplies domain structure but does not create a PINN.

Applying a conventional PINN directly would require an agreed governing formulation, state fields, boundary conditions, and sufficient spatial or temporal observations. This database predominantly contains tabular experimental summaries and endpoint outcomes, not common-domain field solutions. The Rivera/Jose paper also notes that the theoretical framework remains incomplete for direct physics-informed deployment [4]. Therefore, the immediate opportunity is more modest: enforce nonnegative FSR, use dimensionally consistent representations, penalise known inadmissible combinations, model regime-specific monotonicity where defensible, or hybridise reduced-order heat-transfer relations with learned residuals. Such constraints must still be evaluated on held-out sources; physical plausibility alone does not guarantee empirical transfer.

5.2.4 Spacecraft fire-safety context

Guibaud and colleagues identify fire as a major concern for inhabited exploration, particularly as missions move farther from immediate Earth support and involve different atmospheres, gravity levels, confinement, and operational constraints [1]. In that setting, ignition and flame spread are not abstract prediction targets. They contribute to material selection, wire and component design, experiment prioritisation, contingency planning, and the evaluation of how a local ignition might develop.

The present findings add a data-governance qualification to that motivation. Historical evidence can support useful interpolation, but mission-relevant environments may resemble a new source domain rather than another row from a known campaign. Lunar and Martian partial gravity, altered oxygen and pressure, aging materials, realistic ventilation, and integrated spacecraft geometries may lie in sparsely observed combinations. The correct literature-level conclusion is therefore not that machine learning replaces fire testing, but that it can organise heterogeneous evidence and identify where that evidence is or is not transferable.

5.3 Real-World Applications (approximately 30%)

5.3.1 Screening and prioritisation

The most defensible near-term application is screening. Given candidate conditions in the documented schema, the ignition artifact can rank cases by predicted ignition probability, and the FSR artifact can estimate spread rate for records sufficiently similar to the training support. This can reduce a large design space to a smaller set requiring detailed simulation or experiment. A screening tool can, for example, flag material–geometry–atmosphere combinations whose predictions and uncertainty suggest elevated concern, or prioritise boundary cases where models disagree.

Such use should be explicitly asymmetric. A high-risk prediction may justify additional investigation; a low-risk prediction should not by itself waive testing. False reassurance is particularly consequential in spacecraft fire safety. Threshold selection must therefore follow an operational loss function rather than a generic metric. If missing a potentially ignitable case is costly, a sensitivity-oriented threshold may be appropriate, accepting more

false alarms. If scarce test capacity is the binding constraint, a more specific threshold may be chosen, but the resulting missed-case risk must be quantified. The repository deliberately stores MCC-, F1-, balanced-accuracy-, and Youden- J -oriented thresholds rather than declaring one universally correct.

FSR screening should rely primarily on RMSE and MAE in physical units, not MAPE near zero. Predictions should be accompanied by a domain flag and an interval or empirical error summary. The grouped champion’s mean RMSE of 0.0122 ms^{-1} , with substantial fold variation, is large relative to many low-spread observations. Ranking may therefore be more reliable than precise point estimation in underrepresented regimes. Where a decision depends on a narrow FSR boundary, confirmatory testing is necessary.

5.3.2 Experiment design and database curation

A second application is directing new data collection. Per-paper errors can identify campaigns that are systematically difficult to predict. Disagreement between random and grouped performance identifies fields that may encode source-specific rather than transferable information. Explainability, used cautiously, can nominate influential variables whose measurement completeness deserves improvement. For example, the repeated importance of fuel pyrolysis temperature, ignition time, thermal diffusion time, oxygen, flow, and gas properties suggests that missing or inconsistently defined values in these fields can be costly.

The tool can also support coverage analysis before an experiment. A proposed test condition can be compared with ranges, categories, and combinations represented in training. A case in a familiar oxygen–pressure range but a new material–geometry–gravity combination should be labelled out of distribution even if each marginal value is individually familiar. This distinction encourages experiments that fill joint gaps rather than merely adding dense repetitions to an already dominant campaign.

Partial-gravity testing is an especially strong candidate for prioritisation. Data collected at lunar- or Martian-relevant gravity, with controlled atmosphere and flow, would test whether a numeric gravity feature transfers smoothly or whether regime-specific dynamics dominate. Because spaceflight experiments are expensive, model uncertainty and coverage can help allocate them, but the acquisition strategy should reserve samples for prospective validation rather than using every new result for immediate retraining.

5.3.3 Material and design decision support

Within a formal engineering workflow, predictions can be one line of evidence in a multi-criteria decision analysis. Candidate materials may be evaluated on ignition propensity and expected spread alongside toxicity, smoke, mechanical performance, electrical function, mass, manufacturability, aging, and compatibility. The models can reveal trade-offs: one material may score lower for ignition under represented conditions yet have uncertain spread behaviour in a new geometry. Presenting both tasks separately is essential because ignition probability and post-ignition spread rate are related but not interchangeable safety endpoints.

A suitable report for each candidate would contain: the input record and units; missing or imputed fields; the model and data fingerprint; interpolation and grouped evidence; the selected threshold policy; predicted probability or FSR; uncertainty or fold-based error context; nearest supported regimes; out-of-distribution warnings; and known limitations. It should also record whether the recommendation is merely “test first”, “requires expert review”, or “within represented screening envelope”. Raw probabilities without this provenance invite overinterpretation.

Human review remains central. A combustion specialist should assess whether the implied trend is physically credible and whether the case violates assumptions hidden by the tabular representation. A materials or systems engineer should evaluate integration effects absent from the database. A safety authority should determine the acceptable evidence and margins. Model output is an aid to these roles, not an automated disposition.

5.3.4 Uncertainty, abstention, and out-of-distribution use

The current repeated-fold standard deviations and cluster-bootstrap intervals provide evaluation uncertainty, not a complete case-specific predictive distribution. Operational use needs both aleatoric uncertainty, arising from experimental variability, and epistemic uncertainty, arising from sparse support and model uncertainty. An ensemble across seeds, folds, or model families could estimate disagreement. Conformal methods could provide empirical intervals or prediction sets if calibrated under a defensible exchangeability assumption. Neither removes distribution shift; grouped or source-conditional calibration remains necessary.

Abstention is a safety feature. The system should decline to issue a routine prediction when required fields are absent, categorical levels are unseen, the query is distant from training support, model portfolios disagree substantially, or a physically admissible range is violated. Out-of-distribution detection should consider mixed data types and joint structure. A simple min–max range check is insufficient because impossible or unsupported combinations can lie inside all marginal ranges.

For ignition, calibration must be examined under held-out papers. A nominal probability of 0.8 is operationally meaningful only if similar grouped predictions ignite at approximately that rate, within uncertainty. For FSR, an interval should be checked for coverage by paper and regime, not only pooled rows. Performance dashboards should expose the worst papers and regimes rather than average away failure pockets.

5.3.5 No replacement for certification or qualification

These artifacts are not certification models. They have not undergone prospective external validation, formal software qualification, independent verification and validation, uncertainty qualification for a specified design envelope, or acceptance by a regulatory or mission assurance authority. The database is observational and heterogeneous; source-held-out performance is moderate; and the models do not enforce governing physics. No model output should replace required material flammability tests, wire qualification, hazard analysis, engineering judgement, or certification evidence.

This limitation does not make the models unusable. Screening and decision support can be valuable precisely because they precede authoritative testing. The proper claim is that the models summarise existing evidence and can prioritise resources under documented uncertainty. The improper claim is that a predicted “no ignition” or low FSR establishes compliance.

Deployment would require version control and auditability. The repository already records data hashes, feature manifests, split assignments, selected hyperparameters, model cards, and out-of-fold predictions. An operational service should additionally lock the software environment, validate units and schemas, log every query and response, monitor drift, and trigger review when the database or artifact changes. Evaluation artifacts and refitted artifacts must remain clearly separated: the former support performance claims; the latter generate predictions.

5.3.6 A staged deployment concept

A cautious implementation can proceed in stages. In an offline research stage, users inspect leaderboards, paper-level errors, learning curves, calibration, and explanations without affecting decisions. In a shadow stage, predictions accompany conventional assessments but are not used to change them; discrepancies are adjudicated and logged. In a controlled screening stage, outputs may prioritise tests within a predeclared envelope, with mandatory abstention rules and expert sign-off. Only after external prospective evidence could a broader role be considered.

Success criteria should be decision-centred. A useful screener is not merely one with a high ROC–AUC; it reduces testing burden or identifies hazardous cases without exceeding a tolerated miss rate, while remaining calibrated across sources. A useful regressor does not merely minimise pooled RMSE; it provides intervals that cover new-paper observations and rankings that support experimental prioritisation. These criteria align model development with fire-safety outcomes rather than leaderboard optimisation.

5.4 Future Work (approximately 10%)

The immediate priority is external, prospective validation. New experiments should be registered before outcomes are observed, processed through a frozen artifact, and retained as an untouched external test set. Validation should report both pooled and source-specific performance, calibration, abstention frequency, and errors by material, geometry, atmosphere, and gravity regime. Only after this evaluation should the new records enter training.

More partial-gravity data are needed. Purposeful lunar- and Martian-gravity campaigns should vary oxygen, pressure, flow, material, and geometry sufficiently to distinguish gravity effects from source and apparatus effects. Replicate measurements and harmonised uncertainty reporting would separate experimental variability from domain shift.

Uncertainty and out-of-distribution detection should become first-class outputs. Paper-aware ensembles, grouped conformal calibration where assumptions are supportable, mixed-type distance or density models, and explicit abstention thresholds should be evaluated prospectively. Coverage should be measured by source and regime, not only globally.

The database would benefit from a shared ontology: canonical material identifiers; geometry and orientation definitions; units; uncertainty fields; missingness reasons; facility, rig, and diagnostic metadata; and controlled vocabularies for ignition and extinction outcomes. Better provenance would allow laboratory- or facility-held-out validation and reduce accidental source fingerprints.

Finally, physically constrained models should be tested without assuming that a full PINN is immediately appropriate. Useful steps include nonnegative FSR outputs, dimensionally consistent features, regime-aware monotonic constraints, conservation-inspired penalties, and hybrids that learn corrections to reduced-order flame-spread relations. Fully physics-informed formulations become credible only when governing equations, boundary conditions, and compatible field data are available [2, 3]. Every proposed constraint must be assessed under the same paper-disjoint and prospective protocols used to test statistical models.

6 Conclusion

This research asked whether heterogeneous literature data can support reliable machine-learning prediction of material ignition and flame spread rate for fire-safe spacecraft design, especially when prediction is required for an experimental source that the model has not previously seen. The precise answer is conditional. The database supports strong interpolation among rows from represented papers, but only moderate and substantially less stable generalisation to held-out papers. It is therefore suitable as the basis of a traceable screening and research-prioritisation tool, but it does not yet support universal prediction, causal claims, or replacement of qualification and certification testing.

For FSR regression, 2,531 usable observations from 66 canonical papers were evaluated with frozen random-row and paper-grouped holdouts. The selected random-forest baseline achieved $R^2 = 0.879 \pm 0.018$, RMSE = $0.00626 \pm 0.00047 \text{ m s}^{-1}$, and MAE = $0.00210 \pm 0.00017 \text{ m s}^{-1}$ under random-row interpolation. When complete papers were withheld, the same model achieved $R^2 = 0.458 \pm 0.222$, RMSE = $0.01221 \pm 0.00412 \text{ m s}^{-1}$, and MAE = $0.00482 \pm 0.00114 \text{ m s}^{-1}$. For ignition classification, 4,487 deduplicated observations from 87 papers were evaluated. The paper-square-root-weighted XGBoost interpolation champion achieved ROC-AUC = 0.908 ± 0.010 and PR-AUC = 0.969 ± 0.004 ; the unweighted XGBoost grouped champion achieved ROC-AUC = 0.718 ± 0.044 , PR-AUC = 0.877 ± 0.030 , and Brier score = 0.179 ± 0.017 .

The gap between these protocols is the primary result. Random cross-validation answers whether a model can recover nearby conditions in campaigns it has partly observed. Paper-grouped validation asks whether it can transfer across a publication boundary that bundles apparatus, laboratory practice, sampled regime, and reporting convention. The latter is the more relevant internal evidence for a new campaign, although it is not a substitute for external validation. Quoting only the random-split scores would materially overstate readiness.

The model comparison also answers the algorithmic part of the question. Tree ensembles are well matched to these mixed, sparse, nonlinear tabular data. XGBoost led ignition prediction, while random forest was selected for FSR; KNN, SVM/SVR, decision-tree, and MLP candidates supplied useful baselines but did not displace the grouped champions. This is consistent with tabular-learning literature, while also demonstrating that XGBoost is not universally best [5]. The wire-specific $R^2 > 0.9$ findings of Rivera/Jose and colleagues remain compatible with the present results because they concern random cross-validation within a narrower electrical-wire database, not paper-disjoint transfer [4].

Feature attributions identified physically plausible variables, including oxygen, flow, thermal diffusion time, fuel pyrolysis temperature, ignition time, and gas or material properties. These findings are diagnostic rather than causal. SHAP decomposes the fitted model; it does not identify the effect of an intervention. Correlated derived variables, uneven missingness, and campaign confounding prevent causal interpretation.

Accordingly, the practical contribution is a reproducible decision-support framework. It freezes splits, isolates tuning within outer training data, verifies paper disjointness, records data fingerprints and feature manifests, reports repeated-fold and paper-level uncertainty, and distinguishes unbiased evaluation from full-data refitting. Within a documented applicability domain, its predictions can rank candidate tests, identify poorly covered regimes, and focus expert attention. They must be accompanied by uncertainty, out-of-distribution checks, threshold rationale, and explicit abstention.

The work therefore establishes a useful but bounded capability: literature-scale machine learning can extract transferable signal for spacecraft fire-safety screening, yet database heterogeneity and source shift remain the dominant obstacles. The path to stronger evidence is not a larger headline score on mixed rows. It is prospective external validation, targeted partial-gravity experiments, a harmonised ontology, uncertainty-aware abstention, and physically constrained models evaluated against genuinely new sources. Until those steps are completed, the system should inform experiments and engineering judgement, never replace them.

A Technical Appendices

A.1 Feature dictionary

The deployed manifests are the authoritative machine-readable feature definitions. Tables A.1 and A.2 provide a compact human-readable dictionary. Units are encoded in names where available; otherwise they follow the database schema. Derived quantities are predictors, not independent measurements.

Table A.1: Ignition feature dictionary (**table-data-v5**; 32 numeric and five categorical features).

Group	Features and interpretation
Fuel properties	<code>fuel_density_kg_m3</code> , <code>fuel_k_w_mk</code> , <code>fuel_cp_j_kgk</code> , <code>fuel_pyrolysis_t_k</code> , <code>fuel_alpha_m2_s</code> , and <code>fuel_volumetric_heat_capacity_j_m3k</code> : density, conductivity, specific heat, pyrolysis temperature, diffusivity, and volumetric heat capacity.
Core properties	<code>core_density_kg_m3</code> , <code>core_k_w_mk</code> , <code>core_cp_j_kgk</code> , and <code>core_volumetric_heat_capacity_j_m3k</code> .
Gas and transport	<code>oxygen</code> , <code>gas_molar_mass</code> , <code>gas_cp_j_kgk</code> , <code>gas_k_w_mk</code> , <code>gas_density_kg_m3</code> , <code>gas_alpha_m2_s</code> , <code>gas_nu_m2_s</code> , <code>reynolds</code> , <code>peclet</code> , <code>prandtl</code> , and <code>thermal_diffusion_time_s</code> .
Operating conditions	<code>pressure_pa</code> , <code>signed_flow_velocity_m_s</code> , <code>flow_speed_abs_m_s</code> , <code>gravity_g</code> , <code>ignition_power_w</code> , and <code>ignition_time_s</code> .
Geometry	<code>half_thickness_m</code> , <code>characteristic_length_m</code> , and <code>sample_dim_1_m</code> through <code>sample_dim_3_m</code> .
Categorical regime	<code>geometry_cat</code> (wire, flat, cylindrical, spherical), <code>diluent_cat</code> , <code>flow_direction_cat</code> (coflow, counterflow, quiescent), <code>gravity_regime_cat</code> (microgravity, partial, Earth, hyper/unknown), and <code>ignition_method_cat</code> .

Table A.2: FSR feature dictionary (**fsr-data-v2**; exact availability follows the artifact manifest).

Group	Manifest fields
Geometry	Sample dimensions, <code>half_thickness</code> , <code>characteristic_length_m</code> , and categorical <code>Geometry of Sample</code> .
Fuel properties	<code>fuel_density_kg_m3</code> , <code>fuel_k_W_mK</code> , <code>fuel_cp_J_kgK</code> , <code>fuel_pyrolysis_T_K</code> , <code>fuel_alpha_m2_s</code> , and <code>fuel_volumetric_heat_capacity_J_m3K</code> .
Core properties	<code>core_density_kg_m3</code> , <code>core_k_W_mK</code> , <code>core_cp_J_kgK</code> , and <code>core_volumetric_heat_capacity_J_m3K</code> .
Gas and transport	Oxygen Concentration, <code>gas_M</code> , <code>gas_cp_mass</code> , <code>gas_k</code> , <code>gas_density_kg_m3</code> , <code>gas_alpha_m2_s</code> , <code>gas_nu_m2_s</code> , <code>Reynolds</code> , <code>Peclet</code> , <code>Prandtl</code> , and <code>thermal_diffusion_time</code> .
Conditions and apparatus	Pressure (Pa), Flow Velocity (m/s), Gravity, Ignition Power (W), Ignition Time (s), categorical diluent, Ignition Method, and antecedent categorical <code>Ignition</code> .

Explicit source identifiers, DOI/citation fields, duplicate FSR targets, and post-outcome variables such as flame length, heat-release rate, smoke, and extinction indicators are excluded from predictors. Paper identity is retained only for grouping and weighting. Numeric parsing accepts cells beginning with a number; strings such as N2 and CO2 remain categorical. For non-XGBoost models, numeric missing values are median-imputed and categorical values receive a constant unknown level; XGBoost may retain native numeric missingness.

A.2 Metric definitions

For observed FSR y_i , prediction $\hat{y}_i$, sample mean $\bar{y}$, and n test observations,

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (\text{A.1})$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (\text{A.2})$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (\text{A.3})$$

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i), \quad (\text{A.4})$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (\text{A.5})$$

NRMSE is the repository’s normalised RMSE output. MAPE is unstable when y_i approaches zero and is secondary to RMSE and MAE for this FSR distribution.

For ignition probability $\hat{p}_i$, the Brier score is

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (\text{A.6})$$

ROC–AUC measures ranking of positives above negatives across thresholds. PR–AUC summarises precision–recall performance and must be read in light of class prevalence. Given threshold τ ,

$$\hat{y}_i = \mathbf{1}[\hat{p}_i \geq \tau]. \quad (\text{A.7})$$

Sensitivity is $\text{TP}/(\text{TP} + \text{FN})$, specificity is $\text{TN}/(\text{TN} + \text{FP})$, precision is $\text{TP}/(\text{TP} + \text{FP})$, and balanced accuracy is the mean of sensitivity and specificity. The Matthews correlation coefficient is

$$\text{MCC} = \frac{\text{TPTN} - \text{FPFN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (\text{A.8})$$

Thresholds optimising MCC, F1, balanced accuracy, and Youden’s J are selected from inner out-of-fold predictions, never from the corresponding outer test labels.

A.3 Reproducibility workflow

The reproducibility contract is:

1. Load the named database and record its SHA-256 fingerprint, data version, parsing decisions, target counts, missingness, and duplicate removal.
2. Canonicalise DOI or citation into `paper_id`; exclude source identifiers and post-outcome leakage from model inputs.
3. Freeze all row and paper-grouped split assignments before model comparison. Assert complete row coverage, no row overlap, and zero paper overlap for grouped protocols.
4. For every candidate and outer fold, fit preprocessing only on the outer training partition. Tune hyperparameters on inner folds; use group-disjoint inner folds when the outer protocol is grouped.
5. Fit the selected inner configuration on the complete outer training partition and score the untouched outer test partition once. For ignition, freeze thresholds from inner out-of-fold probabilities. For FSR bootstrap conditions, augment training only and preserve inherited paper identity.
6. Persist outer predictions, fold and per-paper metrics, search histories, selected hyperparameters, integrity checks, and failure records. Aggregate only complete, valid candidates.
7. Select champions from persisted tables under the versioned policy. The FSR policy prioritises RMSE; the ignition policy prioritises ROC–AUC and requires LOPO robustness for extrapolation.
8. Refit the selected configuration on all labelled data for inference, package the model card and feature manifest, and retain evaluation results separately. A refitted artifact is deployable; outer-fold predictions are the unbiased comparison evidence.

The recorded fingerprints are shown below; random seeds begin at 42 and split files are shared across candidates.

FSR: 6e09a99e4ea69148e69151bd46149be4140d181b2b8294106927ea9499002fc8
 Ignition: d80da0648ca3a7d77d31ae90e53b4e81b2ab28c8ec666ce6cd150ee7544f6

A.4 Traceability matrix

Table A.3: Claim-to-evidence traceability matrix. Paths are relative to the repository root.

Claim	Persisted evidence
FSR dataset has 2,531 target rows from 66 papers	FSRRegressors/results/report/00_overview/dataset_integrity_summary.md.
FSR interpolation and grouped champion scores	FSRRegressors/results/report/README.md, 00_overview/interpolation_leaderboard.md, and extrapolation_leaderboard.md.
FSR generalisation gap and stability	00_overview/generalization_gap.md and metric_stability.md.
Ignition dataset has 4,487 rows, 87 papers, and 3,383/1,104 class counts	IgnitionClassifiers/results/report/dataset_integrity_summary.md.
Ignition interpolation and grouped scores	interpolation_leaderboard.md, extrapolation_leaderboard.md, and interpolation_extrapolation_drop.md.
Threshold trade-offs	IgnitionClassifiers/results/report/champion_threshold_performance.md.
Explanations are model-dependent	Both report directories' SHAP and permutation-importance tables and status JSON files.
Champion identity and full-data artifact provenance	Each task's results/selection.json, artifacts/*_champion/model_card.json, feature manifest, and training-data fingerprint.
No paper leakage and nested tuning contract	FSRRegressors/pipeline.md and IgnitionClassifiers/pipeline.md; machine checks are persisted under results/evaluation.

A.5 Oral-defense reconciliation

The following concise answers reconcile statements that can otherwise appear contradictory.

- Why can the model be called strong if grouped performance is moderate?** It is strong for interpolation and useful but limited for transfer. The report does not collapse those questions into one score.
- Why are Rivera/Jose results above 0.9 while this grouped FSR R^2 is 0.458?** Their headline results concern a narrower wire database under random ten-fold cross-validation; this study's 0.458 holds out entire papers. The protocols, source composition, features, units, and algorithms are not equivalent [4].
- Does paper-grouped validation prove performance on a future spacecraft?** No. It is stronger internal evidence against paper leakage, but future conditions may differ by laboratory, material, facility, gravity regime, and mission environment. Prospective external validation remains necessary.
- Why use XGBoost if random forest wins FSR?** XGBoost is a strong tabular prior and wins ignition, consistent with the literature [5]; model selection is empirical and task-specific. Random forest offered the preferred grouped FSR result in the tested portfolio.
- Does SHAP identify combustion mechanisms?** No. SHAP explains the fitted prediction relative to a background distribution. Correlation, derived features, missingness, and source confounding preclude causal interpretation.
- Did bootstrap augmentation create new experiments?** No. It resampled observed training rows, inherited their paper IDs, and changed weighting. It added no new physical support.
- Can a low predicted risk certify a material?** No. The artifacts support screening and prioritisation only. Qualification, certification, and expert hazard assessment remain authoritative.
- Why not deploy a PINN immediately?** The database mainly contains heterogeneous tabular endpoints, not common-domain fields with agreed governing equations and boundary conditions. Near-term physically constrained or hybrid models are more defensible [3].

A.6 Model portfolios

The executable YAML defines the configured portfolio: 34 FSR candidates (10 XGBoost, seven random forest, six KNN, five MLP, and six SVR) and 26 ignition candidates. The persisted evidence is narrower: the merged FSR manifest contains 25 candidates, each with real-data (rd) and bootstrap-augmented (rd_bt) conditions, while the formal ignition selection record omits MLP despite available shard metrics. The selected champions are `rf_baseline [rd]` for both FSR protocols, `xgb_paper_sqrt` for ignition interpolation, and `xgb_baseline` for ignition extrapolation. These names identify frozen configurations within the persisted comparison pools and do not imply universal superiority outside the recorded schema and validation design.

Table A.4: Configured model portfolios sourced from the repository.

Task	Family	Candidate hypotheses
FSR	XGBoost (10)	Baseline; square-root/inverse paper weights; absolute-error and pseudo-Huber objectives; signed-log($1 + y$) target; paper bagging.
FSR	Random forest (7)	Baseline and paper weights; absolute-error criteria; signed-log target variants.
FSR	KNN (6)	Uniform or distance neighbours, each with no, square-root, or inverse paper weighting.
FSR	MLP (5)	Baseline; paper weighting; signed-log target variants.
FSR	SVR (6)	Linear or RBF kernels, each with no, square-root, or inverse paper weighting.
Ignition	XGBoost (8)	Baseline; paper-only and class-only weights; combined square-root/inverse weights; focal-loss variants.
Ignition	Decision tree (3)	Baseline and combined class plus square-root/inverse paper weighting.
Ignition	KNN (5)	Baseline; weighted variants; distance-neighbour variants.
Ignition	MLP (5)	Baseline; weighted variants; focal-loss variants.
Ignition	SVM (5)	RBF baseline/weighted variants and linear-kernel weighted variants.

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